SRX400 MNHA Secure Branch Design

Validated with GeoIP (ATP Cloud SecIntel), IPsec to a hub, DHCP with lease synchronization, and Juniper Mist

Eric Sandoval, Network Security Architect, HPE · October 2026 (public release 1.0, November 2026) · SRX400 MNHA pair, Junos OS Evolved 25.4X300-D30 · test builds per result in section 2

Summary. This paper documents a validated branch design on an **SRX400 Multinode High Availability (MNHA) active/backup pair** managed by Juniper Mist. It covers ATP Cloud enrollment, feed download on both nodes, per-tenant **source NAT**, an **IPsec overlay to a remote vSRX hub** from each node, and a **DHCP server with lease synchronization** between the nodes. The traffic results in section 6 were measured from a tenant host: traffic to **Germany (DE)** was denied (on-box test policies) and traffic to the **United States (US)** permitted on whichever node was active, before and after an SRG1 failover; traffic to the hub site crossed IPsec from whichever node was active, with the hub's return route following the active node; and the host kept its DHCP address throughout. Section 2 lists the build each result was tested on, and section 9 lists what is not yet tested.

Mist design rule: configure each SRX400 MNHA node in its **own Hub Profile** (Organization → WAN → Hub Profiles), assigned to that one device — **not in a WAN Edge Template**. Each node's values (MNHA IDs, addresses, SRG1 priority) then sit in its own profile, where the Mist UI shows and edits them directly. Both SRX400 nodes remain overlay **spokes**; the profile name does not change their role. See section 4.2.

Contents

1. Topology

2. Validated outcomes

3. How it works

- 3.1 ATP Cloud enrollment
- 3.2 GeoIP feed and policy
- 3.3 MNHA: both nodes stay current
- 3.4 Source NAT
- 3.5 Note: SecIntel over fxp0
- 3.6 DHCP server with MNHA lease synchronization
- 3.7 IPsec to the hub: BGP over IPsec
- 3.8 IPsec to the hub: static over IPsec
- 3.9 ISP failure behavior
- 3.10 LAN link failure behavior

4. Mist-managed deployment

- 4.1 What Mist manages
- 4.2 Use Hub Profiles, not a WAN Edge Template
- 4.3 Hub Profile settings (node 2)
- 4.4 Additional CLI: MNHA block (node 2)
- 4.5 Hub Profile for the vSRX hub

5. Configuration

- 5.1 GeoIP object and policy (both nodes)
- 5.2 ATP Cloud / SecIntel (written by enrollment on each node)
- 5.3 Source NAT (egress node)
- 5.4 Cross-node routing and policy (both nodes)
- 5.5 DHCP server with Active Leasequery (node 2 shown)
- 5.6 IPsec to the hub (node 2 shown)
- 5.7 Static over IPsec (alternative to 5.6 BGP)

6. Results

- 6.1 GeoIP with live traffic, before and after failover
- 6.2 Enrollment, feeds and DHCP
- 6.3 IPsec to the hub, before and after failover
- 6.4 ISP 1 failure: BGP over IPsec vs static over IPsec
- 6.5 LAN link failure on the active node

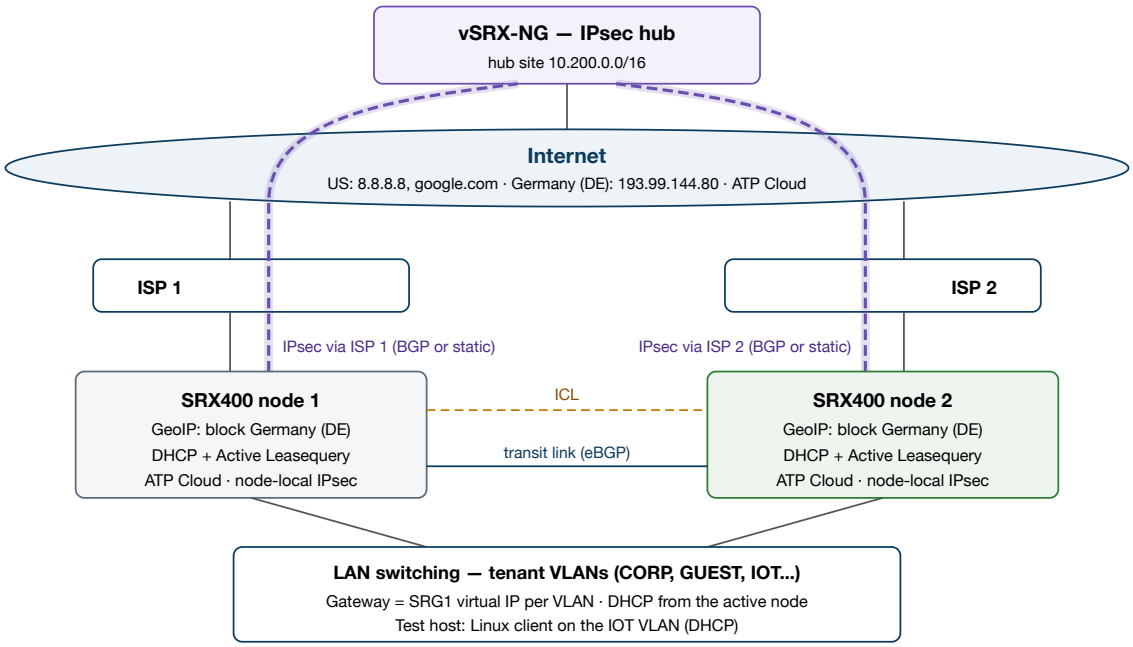
7. Verification checklist

8. Deployment order

9. Not yet tested

10. Reference configurations

1. Topology



The SRG1-active node owns the tenant gateway virtual IPs and answers DHCP. Each tenant exits through its preferred ISP; a tenant whose ISP is on the peer node crosses the transit link (ge-0/0/5, zone `ICD`). Both nodes carry the GeoIP object and the IOT deny policy on both exit paths (IOT → WAN and IOT → ICD), so the block applies on whichever node is active. The live test, with on-box test policies, covered node 2 IOT → ICD and node 1 IOT → wan (section 6.1). Each node also runs its own IPsec tunnel to the vSRX hub; IOT traffic to the hub site is carried over the tunnel of the active node.

2. Validated outcomes

Outcome	Result	Tested on
Traffic to Germany (DE) denied by the GeoIP policy; US permitted — node 2 active	**PASS**	Hub Profile build, 15 Oct (on-box test policies, 6.1)
Same result after SRG1 failover — node 1 active	**PASS**	Hub Profile build, 15 Oct (on-box test policies, 6.1)
IOT test host kept its DHCP address across one failover (lease still valid; renew not tested)	**PASS**	Hub Profile build, 15 Oct; lease sync on this build not yet validated (section 9)
IPsec tunnel and BGP to the vSRX hub up on both nodes	**PASS**	Hub Profile build, 15 Oct
Traffic to the hub site over IPsec — node 2 active, and node 1 after failover	**PASS**	Hub Profile build, 15 Oct
Hub return route follows the active node within about one second	**PASS**	Hub Profile build, 15 Oct; also WAN Edge Template build, 19 Oct (6.5)
ISP 1 failure on the active node: new flows (pings) to the hub site continue via the peer's tunnel — BGP over IPsec and static over IPsec	**PASS**	Hub Profile build, 15 Oct (static routes added on the box, 6.4)

LAN link failure on the active node: SRG1 moves to the peer (interface monitoring); no packet loss to the hub site through failure, restore and failback	**PASS**	WAN Edge Template build, 19 Oct
ATP Cloud enrollment of each node, by chassis serial	**PASS**	Hub Profile build, 15 Oct
GeoIP feed downloaded on both nodes (DE = 31,029 IPv4 / 27,638 IPv6 entries)	**PASS**	Hub Profile build, 15 Oct
Backup node keeps refreshing feeds, including after failover and failback	**PASS**	Hub Profile build, 15 Oct
Note: SecIntel feeds also download over fxp0 (`mgmt_junos`)	**PASS**	Hub Profile build, 15 Oct; node 1, on-box test change
DHCP lease held on both nodes; same IP after failover, failback and node reboot	**PASS**	Hub Profiles, 5 Oct (earlier configuration)

Test builds.
Hub Profile build, 15 Oct 2026 = the design in sections 4 and 5.
WAN Edge Template build, 19 Oct 2026 = a separate build of the same pair, with Mist's native MNHA settings in a WAN Edge Template and per-node values set through the Mist API. It differs from sections 4 and 5 in the ICL, the liveness multiplier, the SRG layout and the routing: the WAN in inet.0 with no WAN-VR, eBGP over ge-0/0/5 in VR `lan` (BFD 1000 ms × 3), and source NAT on tenant → ICD and ICD → overlay traffic. Its configuration was Mist's render, applied on the box with the devices unmanaged in Mist. It was built on 16–17 Oct; the LAN-link test ran on 19 Oct.
Hub Profiles, 5 Oct 2026 = an earlier configuration of the same Hub Profiles. The DHCP client was an EX2300 switch on CORP; section 3.6 lists the `lan` ALQ peer each run used.
Both nodes ran Junos OS Evolved 25.4X300-D30 in all three builds. Section 6 lists the on-box test changes each result used. Section 9 lists what is not yet tested.

Some names, the tenant networks (names, VLANs and subnets) and the hub-site addresses (10.200.0.0/16) were changed for publication.

3. How it works

3.1 ATP Cloud enrollment

Each node is enrolled on its own with the command generated by the ATP Cloud portal (Devices → Enroll), run on the device. The SRX400 is identified and entitled by its **chassis serial**; no license key is installed on the box. Both nodes enrolled at the premium tier and confirmed with "*Device enrolled successfully!*".

```
request services advanced-anti-malware enroll https://<region>.sky.junipersecurity.net/<enroll-path>/<token>.slax
```

Copy the full enrollment command from the ATP Cloud portal.

3.2 GeoIP feed and policy

The GeoIP category is downloaded once a `dynamic-address` object references it. The object for Germany (DE) resolved to 31,029 IPv4 and 27,638 IPv6 entries on both nodes. A deny policy that matches the object, placed above the tenant's Internet permit, blocks the country. In an MNHA branch, add it to every zone pair a tenant can exit through: tenant → WAN and tenant → ICD.

Fail-open. If the feed is empty or not yet downloaded (for example before ATP enrollment), the object matches nothing and the deny policy does not block. Watch the entry count (section 7). Where blocking must fail closed, an alternative design (not

tested here) permits only the allowed countries and denies the rest, at the cost of blocking all traffic while the feed is unavailable.

3.3 MNHA: both nodes stay current

Each node holds its own ATP Cloud connection and downloads feeds independently, so GeoIP data is already current on the backup when it takes over. The backup node kept refreshing feeds after a manual failover and after failback.

3.4 Source NAT

Each tenant zone source-NATs on the WAN of the node that egresses its traffic. Traffic that arrives from the peer over the inter-node transit link (zone `ICD`, ge-0/0/5) is translated by the egress node's `ICD_to_wan` (node 1) or `ICD_to_wan2` (node 2) rule-set. Both nodes are configured for interface NAT (section 5.3). The live Internet test in section 6.1 ran through ISP 1, with a temporary pool address on node 1 in place of interface NAT. As-built interface NAT, Internet egress by ISP 2 and per-node pool NAT have not been tested on the Hub Profile build (section 9).

The SRX400 supports one data SRG (SRG1), so a tenant whose ISP is on the peer node crosses the transit link in steady state, not only during failures. That is fine for the policy-based GeoIP used here; if IDP or AppSecure is added later, review the cross-node inspection path first.

3.5 Note: SecIntel over fxp0

Most deployments send ATP Cloud and SecIntel traffic over the data plane. In the Hub Profile build this traffic used the out-of-band management network through the LAB ONLY inet.0 default (section 4.1). On the WAN Edge Template build (section 2), both nodes enrolled and downloaded feeds through their ISP in inet.0. The data-plane path in this design is not yet tested (section 9). As an on-box test on node 1, SecIntel feeds also downloaded over the out-of-band management port (fxp0; `re0:mgmt-0` on the SRX400), with SecIntel in the management routing instance `mgmt_junos` and no default route in the default instance:

```
set system management-instance
set routing-instances mgmt_junos routing-options static route 0.0.0.0/0 next-hop <mgmt-gateway>
set system name-server <dns> routing-instance mgmt_junos
set services security-intelligence routing-instance mgmt_junos
```

These lines cover SecIntel feed downloads only; the ATP Cloud (AAMW) connection needs its own path to the cloud. `set system management-instance` moves the management port into `mgmt_junos`, and Mist's outbound-ssh session does not use `mgmt_junos`. A node that reaches Mist through its management port loses its Mist connection until the change is reverted. Make the change under `commit confirmed`.

3.6 DHCP server with MNHA lease synchronization

`managed-services dhcp` makes the SRG1-active node the DHCP server. On its own it does not copy bindings to the peer. **Active Leasequery** (ALQ, RFC 7724) does: it copies every binding to the peer over a TCP session (TCP/67) between the nodes.

- **One ALQ peer per DHCP routing instance.** The peer address must be in the same routing instance as the DHCP group and pool. This design has two DHCP routing instances: `lan` (MGMT, CORP, MEDIA, SECURITY, IOT) and `GUEST-VR` (GUEST). Each node peers with the other node's address in each instance: MGMT 10.10.10.2 / .3 in `lan`, GUEST 10.10.30.2 / .3 in `GUEST-VR`.
- **The switch must carry each peer VLAN between the nodes.** That means MGMT (VLAN 10, untagged on the LAN trunk) for `lan` and GUEST (VLAN 30) for `GUEST-VR`.

- **Broadcast replies on D30.2.** 25.4X300-D30.2 needs `dhcp-local-server overrides no-unicast-replies` (section 5.5). With it, OFFER and ACK are broadcast, and a client that does not set the DHCP broadcast flag (for example Linux dhclient) binds. Without it, that client receives no OFFER (tested on node 1, D30.2, 14 Oct). Renewals, and clients that set the flag such as EX switches, bind either way.
- **Failback is manual** (preemption disabled). On the active node, run `request chassis high-availability failover services-redundancy-group 1 peer-id <target>`. First check on the target node: cold sync COMPLETE in `show chassis high-availability information`, and the ALQ peer Connected in both `lan` and `GUEST-VR` (section 7).

Result: 5 Oct 2026, 25.4X300-D30 on both nodes. Both nodes held each lease with the same expiry. After a manual failover, and again after failback, the client's renew was acknowledged with the **same IP**. A renew 1, 4 or 15 s after SRG1 moved kept the IP in 6 of 6 tries, in both directions (and in 6 more with `process-packet-on-backup` added, which this design does not use). A release removed the binding from both nodes within 2 s. After a reboot of the active node, the survivor was ACTIVE within 12 s and kept serving the same address. The rebooted node pulled the bindings back over ALQ before it returned to BACKUP, 369 s after the reboot command.

Test conditions. The 5 Oct run used an earlier revision of these Hub Profiles and an EX2300 DHCP client on CORP, which sets the broadcast flag. The renew-timing runs had the `lan` ALQ peer on MGMT, as built; the failover, failback, release and reboot runs had it on CORP (10.10.20.2 / .3). Not yet validated (section 9): the `no-unicast-replies` line pushed from the Hub Profiles, and failback onto a node rebooted a few minutes earlier.

3.7 IPsec to the hub: BGP over IPsec

Each SRX400 node builds its own IKEv2 tunnel to the remote vSRX hub from its own WAN: certificate (RSA signature) authentication, DH group 14, AES-256-GCM for IKE and ESP, dead-peer detection every 3 s. Tunnels are `node-local`, the MNHA mode for per-node tunnels with dynamic routing, so both tunnels are up at all times, including on the backup. Each node runs iBGP to the hub over its tunnel with BFD (1 s × 3) and advertises the tenant subnet; the SRG1-active node adds local-preference 200, using the active signal route as a policy condition. The hub therefore returns traffic to the active node, and moves to the new active node as soon as SRG1 fails over.

What is synchronized. Firewall sessions are synchronized between the nodes; IPsec security associations are not. Each node negotiates its own tunnel (`node-local`), so after a failover new flows use the new active node's tunnel, steered by the BGP local-preference. Established sessions across a failover are not yet tested (section 9).

3.8 IPsec to the hub: static over IPsec

The same node-local tunnels can carry static routes instead of BGP. Each node points the hub-site prefixes at the hub's tunnel address; the hub points the tenant subnet at the primary node's tunnel address with a floating static (higher preference value) to the other node. When a node's tunnel goes down, IPsec dead-peer detection removes that tunnel's next hop on the hub and the floating static takes over. The static preference is fixed in configuration, whereas BGP sets it from SRG1 state.

3.9 ISP failure behavior

When the active node loses its ISP, SRG1 stays on that node (the LAN is unaffected). The node no longer has its own tunnel, so it forwards the tenant's hub-site traffic across the transit link (ge-0/0/5) to the peer, which sends it out its own tunnel; the hub returns traffic to the peer, which hands it back over the transit link. When the ISP returns, the tunnel re-establishes and the hub moves back to the primary path automatically, with both BGP and static over IPsec. Tested with node 1 active and ISP 1 (section 6.4).

3.10 LAN link failure behavior

SRG1 monitors the LAN interface (`monitor interface ge-0/0/3`). If the active node loses its LAN link, it goes INELIGIBLE (failure event `IF`) and the peer takes over SRG1 in about 0.2 s (device event log): the tenant virtual IPs move, and the active signal route moves with them, so the hub's return path follows the new active node (local-preference 200). When the link returns, the node rejoins as BACKUP. With preemption disabled, moving SRG1 back is a manual command.

Measured on the WAN Edge Template build of 19 Oct 2026 (section 6.5), not on the Hub Profile build of sections 4 and 5. The test used pings to the hub site only; Internet traffic from DHCP clients after the failure is not yet tested (section 9).

4. Mist-managed deployment

4.1 What Mist manages

In this lab the SRX400 pair is managed by Juniper Mist. Each node has its own Hub Profile (section 4.2), so the two nodes receive their own configuration. Mist renders the ports, tenant networks, DHCP, application policies and the IPsec overlay natively. Everything that makes the two nodes work as a pair is carried in each profile's Additional CLI: MNHA, the routing and policies between the nodes, the cross-node NAT, the overlay settings Mist does not render, DHCP lease synchronization and GeoIP.

Rendered natively by Mist	Carried in Additional CLI (per node profile)
• WAN and LAN ports, tenant networks and VLANs, and the node's own address on each network • Tenant networks in routing instance `lan` • DHCP servers and pools • From the application policies and each network's Internet access: tenant → own WAN zone and IOT → overlay security policies, and per-tenant source NAT on the node's own WAN • IPsec overlay to the vSRX hub: certificate-based IKEv2, multipoint `st0`, BGP to the hub (instance `vpn_OrgOverlay`)	• MNHA: SRG1, ICL, MNHA forwarding over the ICL, virtual IPs, signal routes, LAN interface monitoring (section 4.4) • VR routing: `WAN-VR` holding ge-0/0/2 and ge-0/0/5 (LAB ONLY), the `lan` default next-table WAN-VR, `GUEST-VR` for GUEST and its DHCP; Mist APBR removed (`delete apply-groups apbr`) (section 5.4) • Inter-node transit link ge-0/0/5: eBGP between the nodes with BFD, policies `WAN-EXPORT`, `ICD-IMPORT` and `HUB-FROM-OVERLAY`, rib-group `WAN-LEAK` • Zone `ICD` and the cross-node policies: tenant → ICD (`*-xnode`), ICD → WAN (`xnode-out`), ICD → vpn_OrgOverlay (`xnode-iot`) • ICD source NAT: interface NAT, rule-set `ICD_to_wan` on node 1 and `ICD_to_wan2` on node 2 (section 5.3) • Overlay: AS 65000 in vpn_OrgOverlay, `node-local` on the IKE gateway, BGP BFD, active-node local-preference (`IOT-LP` with the `MNHA-ACTIVE` condition), DPD always-send on the Mist-generated IKE gateway (section 5.6) • DHCP `managed-services dhcp` and Active Leasequery, one peer per DHCP routing instance (lan and GUEST-VR) • GeoIP `dynamic-address` and the GEO-BLOCK-DE deny policies (IOT → WAN and IOT → ICD) • ATP Cloud / SecIntel connection lines (section 5.2) • LAB ONLY: inet.0 default to the out-of-band management gateway

- **ATP Cloud enrollment** is done on each device with the ATP Cloud portal command (section 3.1). Enrollment writes PKI, SSL and ATP Cloud configuration on the box.
- **What a Mist push keeps.** Mist pushes the full profile on every change. Configuration written by enrollment stays in place across Mist pushes (Hub Profile build, section 2). An on-box DHCP line (section 5.5) is removed by a later push. Apart from what enrollment writes, keep every change in the profile's Additional CLI.

- **Removed `chassis high-availability` lines stay on the box.** Removing such a line from the Additional CLI stops Mist sending it, but the statement stays in the running configuration of each node until it is deleted on the box. For example, SRG1 `process-packet-on-backup`, once removed from the Additional CLI of both nodes, stays on both nodes (D30.2) until `delete chassis high-availability services-redundancy-group 1 process-packet-on-backup` is committed on each box. After removing any Additional CLI line, check the box for it.
- **Enrollment lines in the profile.** The profiles also carry the section 5.2 lines, which match what enrollment writes. Those lines reference CA profiles and a certificate that only enrollment creates. Pushing them to a node that is not yet enrolled has not been tested (section 9).
- **Set, delete and comment lines only.** Additional CLI accepts `set`, `delete` and `#` comment lines (Figure 4-6). An `insert` line fails the whole push, and Mist shows no reason. Policy order follows line order (section 5.1).
- **Confirm on the box.** The Mist portal status is not final: a commit shown as FAILED can later show COMMITTED, for example after a retry. After each profile change, check `show system commit` for the Mist commit.

4.2 Use Hub Profiles, not a WAN Edge Template

> **Build each SRX400 node in its own Hub Profile** (Organization → WAN → Hub Profiles), assigned to that one device. **Do not use a WAN Edge Template** for the MNHA pair.

The two MNHA nodes must differ: each has its own LAN addresses, ICL and inter-node transit link addresses, MNHA local and peer IDs and SRG1 priority. A WAN Edge Template applies one configuration to the site's gateways. In the separate WAN Edge Template build (section 2), the per-node values were set through the Mist API (node rules and device-level settings). This design keeps each node's values in its own Hub Profile, assigned to that device, where the Mist UI shows them (Figures 4-3 to 4-6). Despite the name, a Hub Profile does not make the SRX400 an overlay hub — the overlay role is set per WAN port, and both SRX400 nodes are **spokes**. The vSRX hub uses a third Hub Profile with the `hub` role.

Hub Profile	Assigned to	Role
node 1 profile	SRX400 node 1	Spoke on the overlay; MNHA local-id 1, SRG1 priority 200
node 2 profile	SRX400 node 2	Spoke on the overlay; MNHA local-id 2, SRG1 priority 100
vSRX hub profile	vSRX-NG	Overlay hub (static WAN address, `role: hub`)

Device-level Additional CLI. In this build node 2 also has a device-level Additional CLI list, identical to its profile's list. A device-level list replaces the profile's list; the two are not merged. An edit made only in the profile does not reach node 2, so keep the two lists identical. Never set the device-level list to an empty list (`[]`) or a partial one: Mist then renders the node from that list alone. A device-level list without the MNHA lines removes them from the box (`allow-ha`, the cross-node NAT, `MNHA-NODE-LOCAL`, the WAN port's WAN-VR membership), except the `chassis high-availability` lines, which stay (section 4.1). Tested with a device-level list holding only a comment (Hub Profile build, section 2): the MNHA peer went down and both nodes went ACTIVE until Junos rolled back the Mist commit (commit confirmed, about 5 minutes). With an empty list, SRG1 moved to the peer and stayed there after the rollback (no preemption). Removing the device-level list, so that the profile's list applies, has not been tested (section 9).

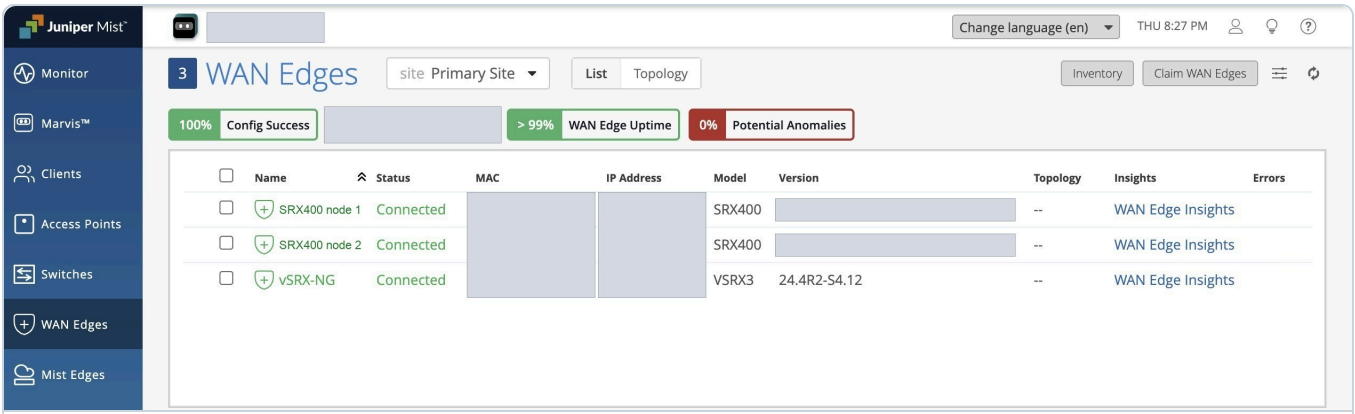


Figure 4-1: WAN Edges inventory: both SRX400 nodes and the vSRX hub connected

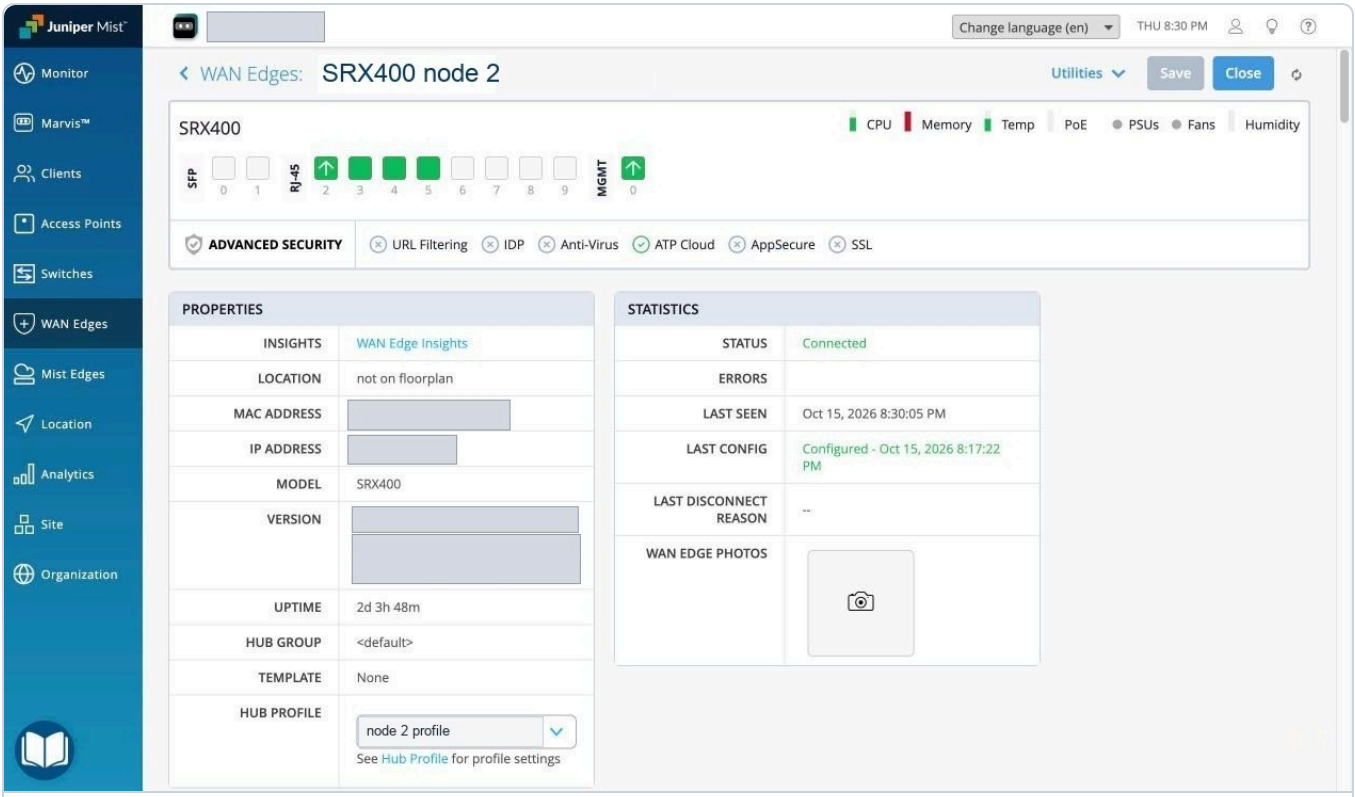


Figure 4-2: SRX400 device page: port status, ATP Cloud enabled under Advanced Security, and the node's own Hub Profile

4.3 Hub Profile settings (node 2)

These are the native settings in node 2's Hub Profile. From them Mist renders the ports, networks, DHCP, security policies, source NAT and the overlay. The Additional CLI then removes Mist's APBR (`delete apply-groups apbr`), so the paths do not steer traffic. VR routing decides which ISP each tenant uses (second table below).

Mist setting	Node 2 value	What it does
WAN port	ge-0/0/2, name **wan2**, broadband, DHCP	Internet uplink to ISP 2. The Additional CLI places ge-0/0/2 in WAN-VR (LAB ONLY)
WAN overlay (Hub to Hub endpoint)	Spoke on the hub's overlay path, BFD profile broadband	Builds the IPsec tunnel to the vSRX hub. The BFD profile is the path setting; BFD on the overlay BGP (1000 ms × 3) is Additional CLI

LAN port	ge-0/0/3: untagged MGMT (VLAN 10); tagged CORP 20, GUEST 30, MEDIA 40, SECURITY 50, IOT 60	Tenant VLAN trunk to the LAN switch
IP config per network	MGMT 10.10.10.**3**/24, CORP 10.10.20.**3**/24, IOT 10.10.60.**3**/24, ... (node 1 uses .2)	The node's own address on each tenant network; the gateway (.1) stays the SRG1 virtual IP
DHCP config	Server on each tenant network, range .10–.200, router .1	Address pools handed out by the active node. The Additional CLI moves the GUEST server and pool to GUEST-VR
VPN access (IOT)	OrgOverlay: routed, no re-advertise to overlay	Advertises the IOT subnet to the hub over the overlay BGP
Traffic steering	Path **wan** = port wan2 · path **OrgOverlay** = the hub overlay path	Named paths the application policies below use. APBR is removed; VR routing decides egress
Application policy **allow-to-hub**	IOT → HUB-SITE-NETS (hub site 10.200.0.0/16), path OrgOverlay	Permits IOT → vpn_OrgOverlay; IOT traffic to the hub site rides IPsec
Application policy **Internet**	All six tenant networks → any, path wan	Permits each tenant → wan2 (`01_Internet`) with source NAT on wan2. VR routing, not this path, sets the ISP a tenant uses

Egress by VR routing (Additional CLI). Each tenant routing instance has a default route `0.0.0.0/0 next-table WAN-VR.inet.0`. Each node also learns its peer's ISP default over ge-0/0/5 (BGP, preference 170), and rib-group WAN-LEAK copies it into lan and GUEST-VR. Where the local default has preference 200, the peer's default wins and the tenant crosses to the peer (section 5.4).

Routing instance	Tenants	Node 1	Node 2
`lan` — ISP 1 side	MGMT, CORP, MEDIA, SECURITY, IOT	Own WAN (ISP 1): local next-table default	Peer's ISP 1 across ge-0/0/5: BGP default (170) beats the local default (preference 200)
`GUEST-VR` — ISP 2 side	GUEST	Peer's ISP 2 across ge-0/0/5: BGP default (170) beats the local default (preference 200)	Own WAN (ISP 2): local next-table default

On the egress node, traffic that crosses ge-0/0/5 is permitted by the Additional CLI policies `xnode-out` (ICD → WAN) and `xnode-iot` (ICD → vpn_OrgOverlay), not by the application policies above. The `ICD_to_wan` (node 1) or `ICD_to_wan2` (node 2) rule-set translates the traffic that leaves by the WAN (interface NAT). VR lan learns the hub-site prefixes from the overlay BGP, so IOT traffic to the hub uses the active node's tunnel. The preference-200 default is the fallback to the local ISP when the peer's default is absent. Internet traffic on that fallback has not been tested (section 9). WAN-VR is LAB ONLY: it keeps the WAN out of inet.0, which holds the out-of-band management default. A Hub Profile build with the WAN in inet.0 has not been tested (section 9).

Node 1 is identical except for its WAN port name (**wan**, so path wan = port wan) and its own addresses (.2 on each tenant network).

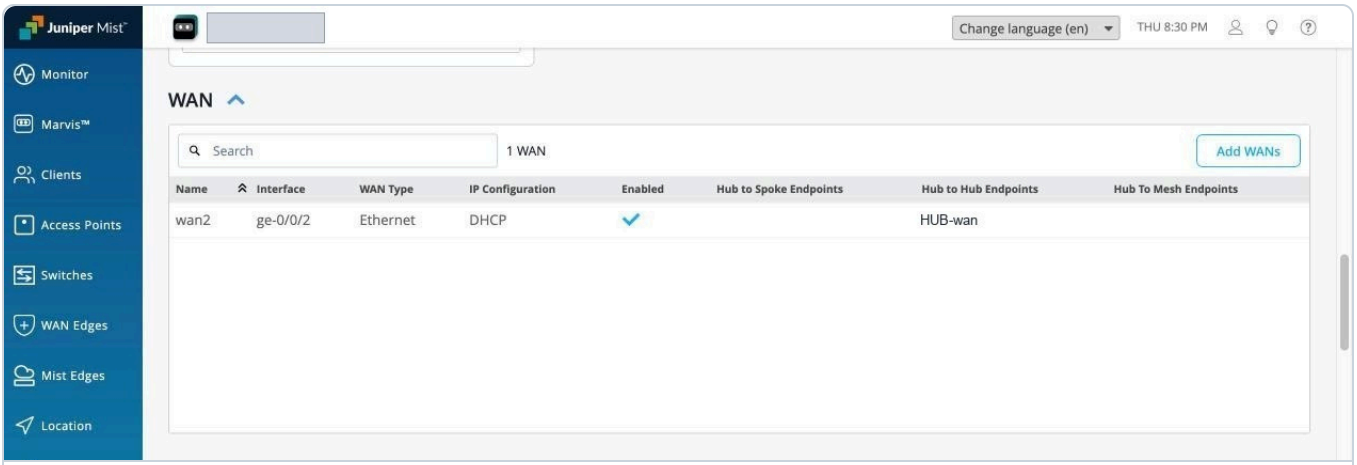


Figure 4-3: Hub Profile: WAN port with its overlay path to the hub

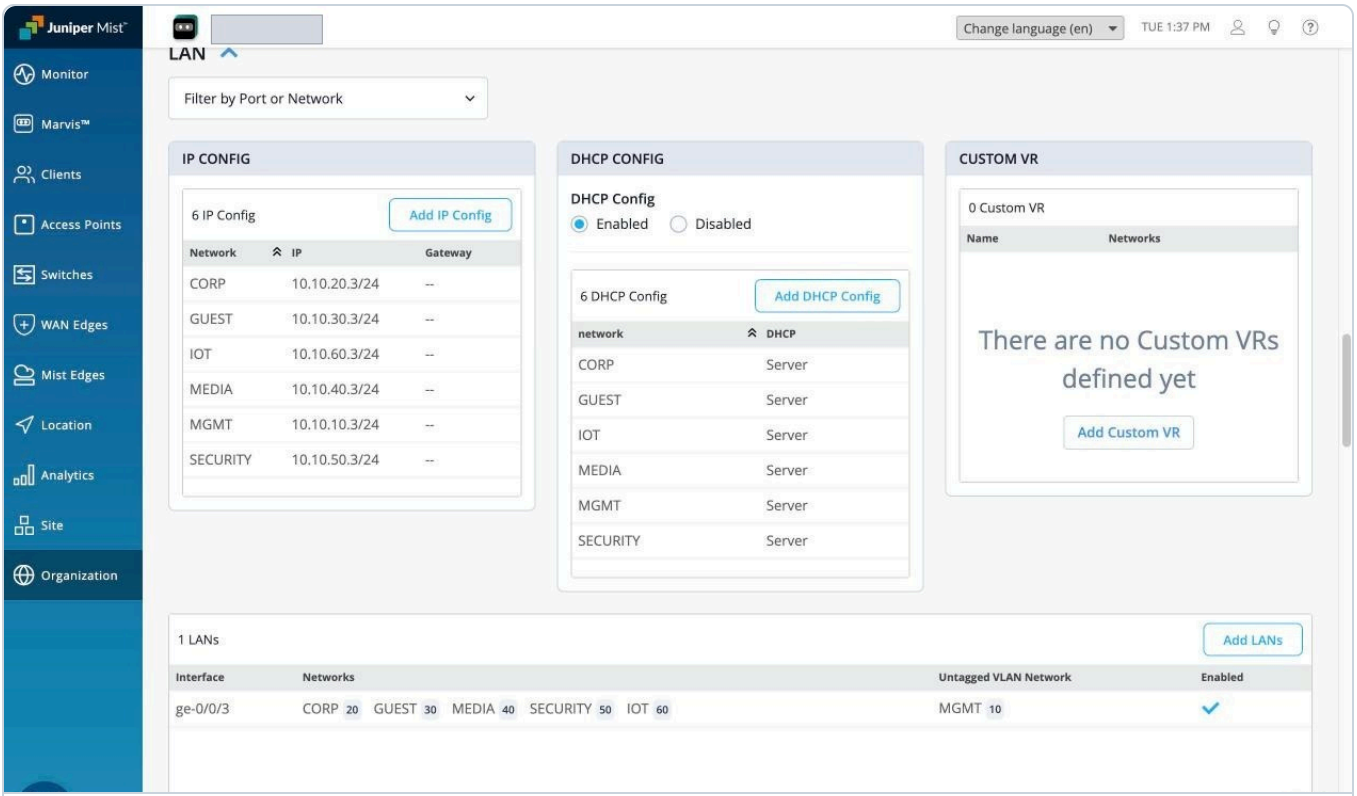


Figure 4-4: Hub Profile: per-node IP configuration on each tenant network, DHCP servers, and the LAN trunk

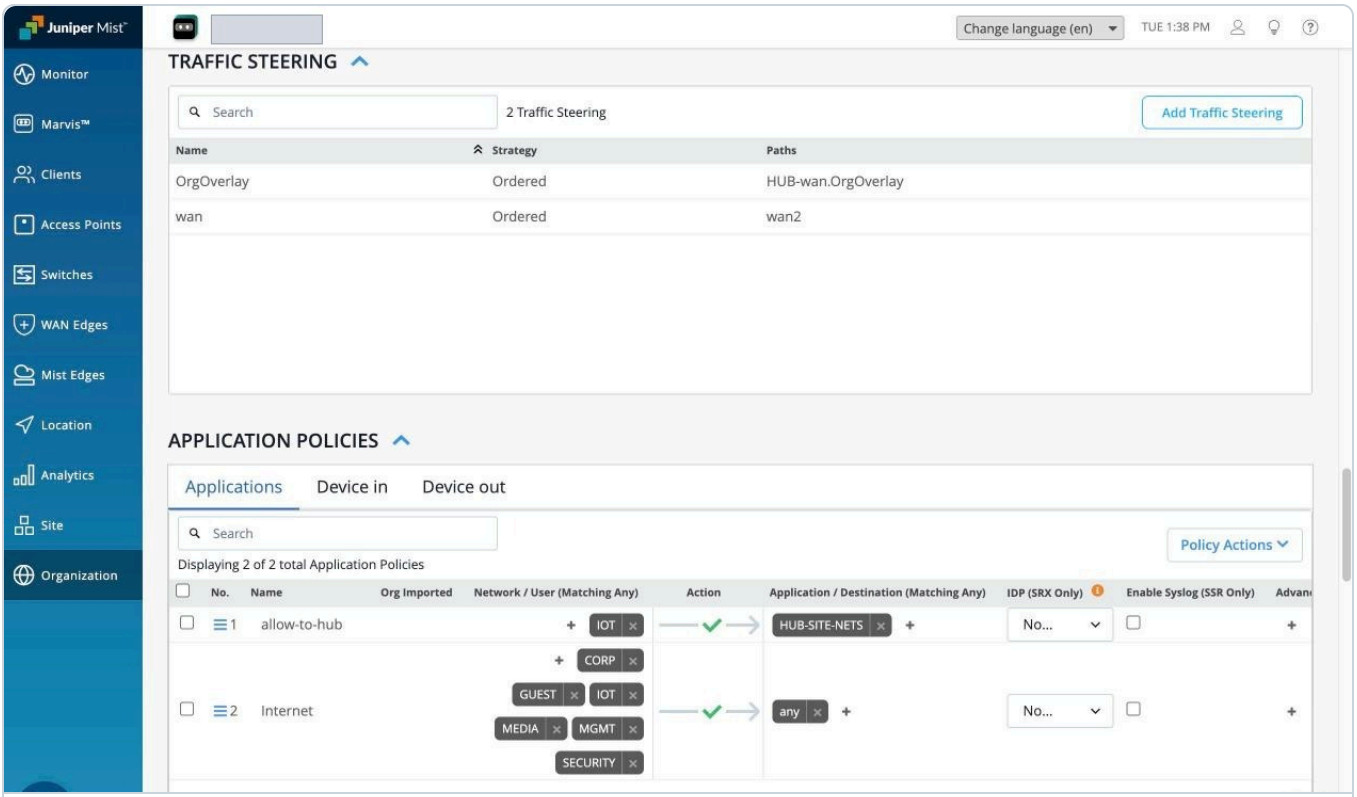


Figure 4-5: Hub Profile: traffic steering paths (wan, OrgOverlay) and application policies (allow-to-hub, Internet)

4.4 Additional CLI: MNHA block (node 2)

This design carries MNHA in each node's Additional CLI: the services redundancy group, the ICL, the MNHA forwarding path and the virtual IPs. Below is node 2's complete MNHA block from its profile. The routing, policies and NAT between the nodes are separate Additional CLI items (section 4.1, section 5.4).

```
# node identity and ICL (ge-0/0/4)
set chassis high-availability local-id 2
set chassis high-availability local-id local-ip 10.255.0.2
set chassis high-availability local-id local-forwarding-ip 10.255.1.2
set chassis high-availability peer-id 1 peer-ip 10.255.0.1
set chassis high-availability peer-id 1 interface ge-0/0/4.0
set chassis high-availability peer-id 1 liveness-detection minimum-interval 1000
set chassis high-availability peer-id 1 liveness-detection multiplier 3
set interfaces ge-0/0/4 unit 0 family inet address 10.255.0.2/30

# MNHA forwarding path on the ICL link
set chassis high-availability peer-id 1 peer-forwarding-ip 10.255.1.1
set chassis high-availability peer-id 1 peer-forwarding-ip interface ge-0/0/4.0
set chassis high-availability peer-id 1 peer-forwarding-ip liveness-detection minimum-interval 1000
set chassis high-availability peer-id 1 peer-forwarding-ip liveness-detection multiplier 3
set interfaces lo0 unit 0 family inet address 10.255.1.2/32
set routing-options static route 10.255.1.1/32 next-hop 10.255.0.1

# zone HA for the ICL and the forwarding addresses
set security zones security-zone HA interfaces ge-0/0/4.0 host-inbound-traffic system-services all
set security zones security-zone HA interfaces ge-0/0/4.0 host-inbound-traffic protocols all
set security zones security-zone HA interfaces lo0.0 host-inbound-traffic system-services all
set security zones security-zone HA interfaces lo0.0 host-inbound-traffic protocols all
set security policies from-zone HA to-zone HA policy allow-ha match source-address any
set security policies from-zone HA to-zone HA policy allow-ha match destination-address any
set security policies from-zone HA to-zone HA policy allow-ha match application any
set security policies from-zone HA to-zone HA policy allow-ha then permit

# SRG1: hybrid, LAN interface monitoring, signal routes, DHCP
set chassis high-availability services-redundancy-group 1 deployment-type hybrid
set chassis high-availability services-redundancy-group 1 peer-id 1
set chassis high-availability services-redundancy-group 1 activeness-priority 100
set chassis high-availability services-redundancy-group 1 monitor interface ge-0/0/3
```

```
set chassis high-availability services-redundancy-group 1 active-signal-route 10.255.3.1
set chassis high-availability services-redundancy-group 1 backup-signal-route 10.255.3.2
set chassis high-availability services-redundancy-group 1 managed-services dhcp

# SRG1 virtual IPs: the tenant gateways (.1) on the LAN units of ge-0/0/3
set chassis high-availability services-redundancy-group 1 virtual-ip 1 ip 10.10.10.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 1 interface ge-0/0/3.0
set chassis high-availability services-redundancy-group 1 virtual-ip 1 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 2 ip 10.10.20.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 2 interface ge-0/0/3.20
set chassis high-availability services-redundancy-group 1 virtual-ip 2 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 3 ip 10.10.30.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 3 interface ge-0/0/3.30
set chassis high-availability services-redundancy-group 1 virtual-ip 3 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 4 ip 10.10.40.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 4 interface ge-0/0/3.40
set chassis high-availability services-redundancy-group 1 virtual-ip 4 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 5 ip 10.10.50.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 5 interface ge-0/0/3.50
set chassis high-availability services-redundancy-group 1 virtual-ip 5 use-virtual-mac
set chassis high-availability services-redundancy-group 1 virtual-ip 6 ip 10.10.60.1/24
set chassis high-availability services-redundancy-group 1 virtual-ip 6 interface ge-0/0/3.60
set chassis high-availability services-redundancy-group 1 virtual-ip 6 use-virtual-mac

# active signal route as a policy condition (overlay local-preference, ge-0/0/5 export)
set policy-options condition MNHA-ACTIVE if-route-exists address-family inet 10.255.3.1/32
set policy-options condition MNHA-ACTIVE if-route-exists address-family inet table inet.0

# node-local on the IKE gateway Mist generates for the overlay
set groups MNHA-NODE-LOCAL security ike gateway <*> node-local
set apply-groups MNHA-NODE-LOCAL

# inter-node transit link (eBGP) - not part of MNHA, see the note below
set interfaces ge-0/0/5 unit 0 family inet address 10.255.2.2/30
```

ICL and MNHA forwarding. ge-0/0/4 is one link. It carries the ICL (10.255.0.x/30), session synchronization and the MNHA forwarding path. `peer-forwarding-ip interface ge-0/0/4.0` puts the forwarding path on the ICL. The forwarding IPs sit on lo0.0, and a static route over the ICL reaches the peer's.

Inter-node transit link (eBGP). ge-0/0/5 (10.255.2.x/30) is a routed link in WAN-VR. It carries tenant traffic whose ISP is on the peer node, hub-site traffic during an ISP failure, and the eBGP session between the nodes (AS 65001 / 65002, BFD 300 ms × 3; section 5.4). MNHA does not track it. Its security zone and BGP group are named `ICD`; the NAT rule-sets are `ICD_to_wan` and `ICD_to_wan2`. It is not the MNHA ICD; MNHA forwarding uses the ICL.

No activeness-probe and no preemption are configured. Moving SRG1 back after a failover is a manual command. ICL failure and the split-brain probe that hybrid mode derives (expected toward virtual IP 1) have not been tested (section 9). A wildcard group applies `node-local` to the IKE gateway Mist generates for the overlay, whatever its generated name.

Node 1 differences. Everything else in the block is identical, including the virtual IPs and signal routes.

- `local-id 1`, `local-ip 10.255.0.1`, `local-forwarding-ip 10.255.1.1`
- `peer-id 2` in every peer-id line and in SRG1; `peer-ip 10.255.0.2`, `peer-forwarding-ip 10.255.1.2`
- SRG1 `activeness-priority 200`
- ge-0/0/4 `10.255.0.1/30`, lo0.0 `10.255.1.1/32`, ge-0/0/5 `10.255.2.1/30`
- `set routing-options static route 10.255.1.2/32 next-hop 10.255.0.2`

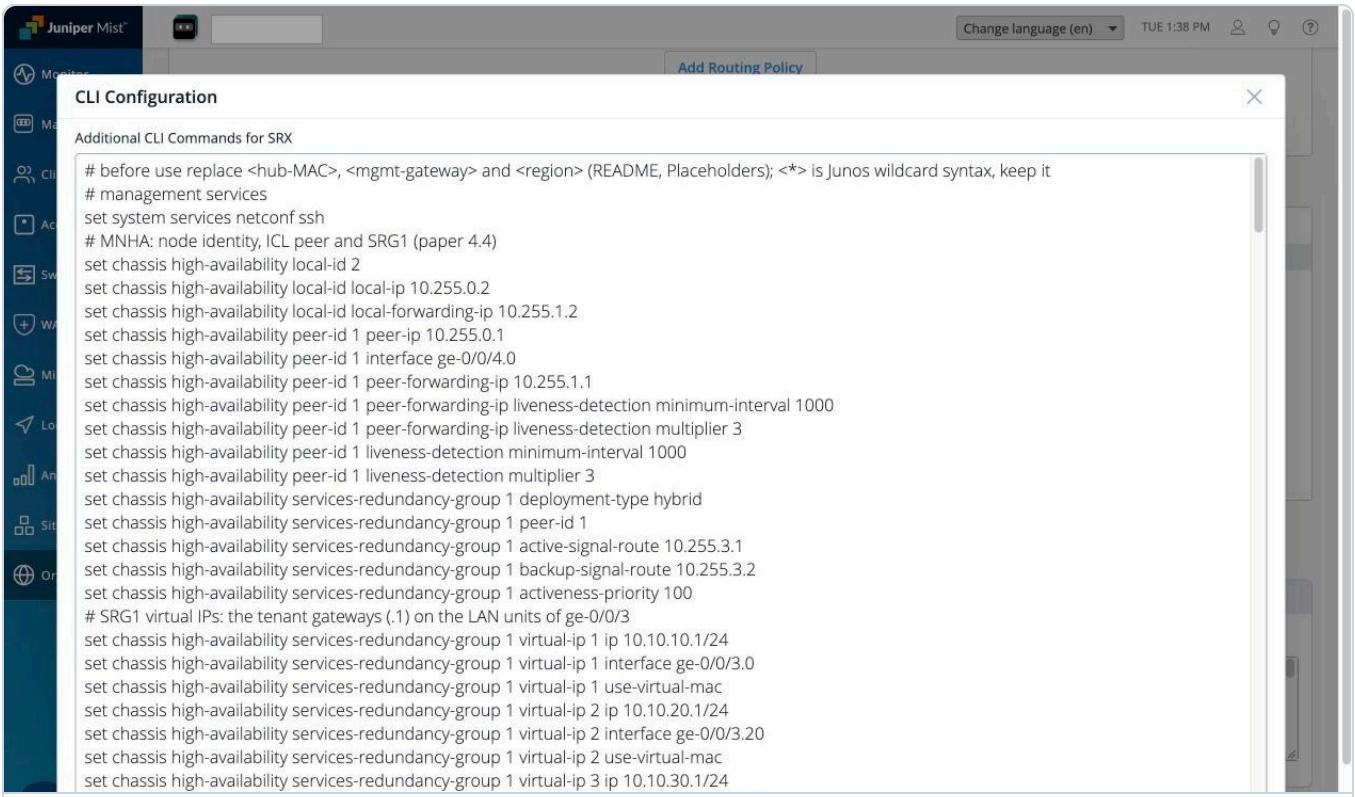


Figure 4-6: Hub Profile: Additional CLI carrying the MNHA statements

4.5 Hub Profile for the vSRX hub

The vSRX hub has its own Hub Profile with the `hub` role.

Mist setting	Hub value	What it does
WAN port	ge-0/0/0, static 172.16.11.2/30, gateway 172.16.11.1	Fixed address the spokes connect to
WAN overlay	Hub role on the overlay path, BFD profile broadband	Terminates both SRX400 tunnels

Mist builds the hub side of the overlay from this profile: certificate-based IKEv2, a multipoint `st0` per path, and the iBGP route reflector toward the spokes in routing instance `vpn_OrgOverlay` (BGP group `OrgOverlay_overlay`, export `OrgOverlay_advertise`). Mist's rendered export matches BGP and OSPF routes only, so it does not advertise the hub's static routes. The hub's Additional CLI adds the hub-site advertisement, an import filter, BGP BFD and DPD always-send.

```
# hub-site prefix behind the hub, advertised to the spokes (lab: discard route + test host on lo0.1)
set routing-instances vpn_OrgOverlay routing-options static route 10.200.0.0/16 discard
set policy-options policy-statement OrgOverlay_advertise term 10_hub_site from protocol static
set policy-options policy-statement OrgOverlay_advertise term 10_hub_site from route-filter 10.200.0.0/16 exact
set policy-options policy-statement OrgOverlay_advertise term 10_hub_site then next-hop self
set policy-options policy-statement OrgOverlay_advertise term 10_hub_site then accept

# LAB ONLY: hub-site test host on the overlay loopback
set interfaces lo0 unit 1 family inet address 10.200.8.1/32

# spokes must not leak MNHA control addresses into the overlay
set policy-options policy-statement OVERLAY-IN term no-mnha from route-filter 10.255.0.0/16 orlonger
set policy-options policy-statement OVERLAY-IN term no-mnha then reject
set policy-options policy-statement OVERLAY-IN term rest then accept
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay import OVERLAY-IN
```

```
# BFD on the overlay iBGP, same timers as the spokes (BGP hold time is 90 s)
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay bfd-liveness-detection minimum-interval 1000
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay bfd-liveness-detection multiplier 3

# DPD always-send on the Mist-generated hub gateway (Mist renders optimized)
delete security ike gateway OrgOverlay_hub_on_wan dead-peer-detection optimized
set security ike gateway OrgOverlay_hub_on_wan dead-peer-detection always-send
```

In the lab, the hub site is the 10.200.0.0/16 discard route plus the test host 10.200.8.1 on `lo0.1`. The `10_hub_site` term exports static routes only. `OVERLAY-IN` rejects only 10.255.0.0/16 and accepts every other spoke route. Mist generates the gateway name `OrgOverlay_hub_on_wan`; take it from the hub's rendered configuration.

The hub's Hub Profile also carries lab-only lines (management, lab routes and a second ISP-side link). They are not part of the design and are not shown.

5. Configuration

5.1 GeoIP object and policy (both nodes)

These lines are in each node's Hub Profile Additional CLI, as built (15 Oct). Policies in a zone pair are evaluated in list order, so the deny must sit above the permit. Mist Additional CLI accepts only `set`, `delete` and `#` comment lines. An `insert` line fails the whole push. The order is therefore set this way:

- **ICD path.** All IOT → ICD policies come from Additional CLI. GEO-BLOCK-DE is listed before IOT-xnode, the cross-node permit, so it is evaluated first. As built, the GEO-BLOCK-DE lines sit just above the IOT-xnode lines of section 5.4 in the list. Put them there; do not append them at the end of the list. A policy's position is set by the first line that creates it. If the GEO-BLOCK-DE lines are appended, the IOT-xnode lines repeated below do not move that policy, and the deny sits below the permit.
- **WAN path.** Mist renders the IOT Internet permit as `01_Internet`. GEO-BLOCK-DE is set after it. Then `01_Internet` is deleted and set again with Mist's exact rendered lines. The re-set policy goes to the end of the list, after the deny.

```
# GEO-DE = all addresses the GeoIP feed maps to Germany (ISO code DE)
set security dynamic-address address-name GEO-DE profile category GeoIP property country string DE

# ICD path (both nodes): GEO-BLOCK-DE listed before IOT-xnode
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE match source-address any
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE match destination-address GEO-DE
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE match application any
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE then deny
set security policies from-zone IOT to-zone ICD policy GEO-BLOCK-DE then count
set security policies from-zone IOT to-zone ICD policy IOT-xnode match source-address any
set security policies from-zone IOT to-zone ICD policy IOT-xnode match destination-address any
set security policies from-zone IOT to-zone ICD policy IOT-xnode match application any
set security policies from-zone IOT to-zone ICD policy IOT-xnode then permit

# WAN path (node 1 shown; node 2 uses zone wan2)
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE match source-address any
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE match destination-address GEO-DE
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE match application any
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE then deny
set security policies from-zone IOT to-zone wan policy GEO-BLOCK-DE then count
# remove Mist's 01_Internet, then re-set it with Mist's exact rendered lines (copy them from the current render)
delete security policies from-zone IOT to-zone wan policy 01_Internet
set security policies from-zone IOT to-zone wan policy 01_Internet match source-address 10-10-60-0_24
set security policies from-zone IOT to-zone wan policy 01_Internet match destination-address any
set security policies from-zone IOT to-zone wan policy 01_Internet match application any
set security policies from-zone IOT to-zone wan policy 01_Internet match dynamic-application any
set security policies from-zone IOT to-zone wan policy 01_Internet then permit
```

Mist generates the name `01_Internet` and the address name `10-10-60-0_24` from the application policy and the IOT network. Copy the re-set lines from Mist's current render of the device. Check them again after any application-policy change. Check the order on both nodes with `show security policies from-zone IOT to-zone wan | no-more` (`wan2`

on node 2) and `show security policies from-zone IOT to-zone ICD | no-more`: GEO-BLOCK-DE has sequence number 1.

The as-built applies GEO-BLOCK-DE to IOT. The WAN zone name is per node: `wan` on node 1, `wan2` on node 2. Another tenant zone needs the same two blocks, with its own xnode policy and its own Mist-rendered Internet policy (not in the as-built).

Test status. The live test in section 6.1 (15 Oct) ran with GEO-BLOCK-DE configured on the box by CLI. The profile lines above were added to the Hub Profiles after that test, the same evening. Both nodes committed them from Mist with GEO-BLOCK-DE first on each zone pair. DE blocked / US allowed has not yet been re-run with the profile-pushed lines (section 9).

5.2 ATP Cloud / SecIntel (written by enrollment on each node)

```
set services advanced-anti-malware connection url https://srxapi-v2.<region>.sky.junipersecurity.net
set services advanced-anti-malware connection authentication tls-profile aamw-ssl
set services security-intelligence url https://cloudfeeds-v2.sky.junipersecurity.net/api/manifest.xml
set services security-intelligence authentication tls-profile aamw-ssl
set services ssl initiation profile aamw-ssl trusted-ca aamw-secintel-ca
set services ssl initiation profile aamw-ssl trusted-ca aamw-cloud-ca
set services ssl initiation profile aamw-ssl client-certificate aamw-srx-cert
set services ssl initiation profile aamw-ssl actions crl disable
```

5.3 Source NAT (egress node)

Both nodes use interface NAT. Traffic is translated to the address of the WAN interface it leaves by: ge-0/0/2, ISP 1 on node 1 and ISP 2 on node 2. No NAT pool is configured in this build. Mist renders the tenant NAT. The NAT for traffic that arrives from the peer over ge-0/0/5 is in Additional CLI, because zone `ICD` is defined there. The rule-set names differ per node because the WAN zone name differs.

Traffic	Configured by	Node 1	Node 2
Tenant zone → local WAN	Mist: Internet application policy and Internet access on each network	One rule-set per tenant zone, `<tenant>_to_wan`, `source-nat interface`	One rule-set per tenant zone, `<tenant>_to_wan2`, `source-nat interface`
ICD → local WAN (traffic from the peer)	Additional CLI	`ICD_to_wan`, rule `xnode`	`ICD_to_wan2`, rule `xnode`

```
# node 2 (Additional CLI)
set security nat source rule-set ICD_to_wan2 from zone ICD
set security nat source rule-set ICD_to_wan2 to zone wan2
set security nat source rule-set ICD_to_wan2 rule xnode match source-address 0.0.0.0/0
set security nat source rule-set ICD_to_wan2 rule xnode then source-nat interface

# node 1 (Additional CLI)
set security nat source rule-set ICD_to_wan from zone ICD
set security nat source rule-set ICD_to_wan to zone wan
set security nat source rule-set ICD_to_wan rule xnode match source-address 0.0.0.0/0
set security nat source rule-set ICD_to_wan rule xnode then source-nat interface
```

The ICD rule-set needs the `ICD → wan|wan2` permit `xnode-out` (section 5.4). `show security nat source rule all` lists every rule-set with its translation hits.

NAT pool. To use a pool in place of the interface address, give each node its own pool, routed to that node by its own ISP. Per-node pool NAT has not been tested (section 9). The 6.1 Internet test used a temporary pool on node 1, in place of interface NAT, for the lab's Internet uplink; that pool is not part of the build. No tenant Internet traffic in the results used the as-built interface NAT.

5.4 Cross-node routing and policy (both nodes)

Routing-instance (VR) routing decides which ISP each tenant uses. Mist renders the tenant networks, the application policies and the tenant NAT. The Additional CLI below does four things. It removes Mist's APBR. It puts the WAN in `WAN-VR` (LAB ONLY). It gives each tenant VR a default into `WAN-VR`. It joins the `WAN-VR` instances of the two nodes with eBGP over ge-0/0/5.

ge-0/0/5 is the inter-node transit link; its zone and BGP group are both named `ICD`. It is a routed link. MNHA forwarding uses the ICL, ge-0/0/4 (section 4.4). Node 2 is shown first, then node 1's differences.

Routing instance (tenants)	Node 1	Node 2
`lan` (MGMT, CORP, MEDIA, SECURITY, IOT)	Static 0/0 next-table `WAN-VR`, default preference (5). Exits ISP 1 locally.	Static 0/0 next-table `WAN-VR`, preference 200. The BGP 0/0 from node 1 (170) is preferred. Exits ISP 1 through node 1.
`GUEST-VR` (GUEST)	Static 0/0 next-table `WAN-VR`, preference 200. The BGP 0/0 from node 2 (170) is preferred. Exits ISP 2 through node 2.	Static 0/0 next-table `WAN-VR`, default preference (5). Exits ISP 2 locally.
`WAN-VR`	DHCP default from ISP 1; BGP 0/0 from node 2	DHCP default from ISP 2; BGP 0/0 from node 1
Hub-site prefixes (10.200.0.0/16)	Each node uses its own tunnel first: `lan` imports these prefixes from `vpn_OrgOverlay` through Mist's `vpn_OrgOverlay_bgp`. `WAN-VR` holds the node's own copy (`HUB-FROM-OVERLAY`, BGP 170) and the peer's copy at preference 175. When a node's own tunnel is down, the `lan` default carries hub traffic across ge-0/0/5 to the peer's tunnel. Tested with node 1 active (section 6.4).	

WAN-VR and the transit link (node 2)

```
set interfaces ge-0/0/5 unit 0 family inet address 10.255.2.2/30
# LAB ONLY: inet.0 keeps the out-of-band management default
set routing-options static route 0.0.0.0/0 qualified-next-hop <mgmt-gateway> preference 1
# LAB ONLY: the WAN (ge-0/0/2, Mist WAN port, DHCP) sits in WAN-VR with the transit link.
# Production build: the WAN stays in inet.0 (not tested).
set routing-instances WAN-VR instance-type virtual-router
set routing-instances WAN-VR interface ge-0/0/2.0
set routing-instances WAN-VR interface ge-0/0/5.0
set routing-instances WAN-VR routing-options router-id 10.255.2.2
set routing-instances WAN-VR routing-options autonomous-system 65002
# tenant subnets: discard at preference 250, exported to the peer only by the SRG1-active node
set routing-instances WAN-VR routing-options static route 10.10.10.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.10.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.20.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.20.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.30.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.30.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.40.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.40.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.50.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.50.0/24 preference 250
set routing-instances WAN-VR routing-options static route 10.10.60.0/24 discard
set routing-instances WAN-VR routing-options static route 10.10.60.0/24 preference 250
# hub-site routes from this node's own tunnel
set routing-instances WAN-VR routing-options instance-import HUB-FROM-OVERLAY
```

Routing policy (identical on both nodes)

```
# hub-site prefixes learned over this node's own tunnel, tagged for export to the peer
set policy-options community HUB-LOCAL members 65000:1000
set policy-options policy-statement HUB-FROM-OVERLAY term hub from instance vpn_OrgOverlay
set policy-options policy-statement HUB-FROM-OVERLAY term hub from protocol bgp
set policy-options policy-statement HUB-FROM-OVERLAY term hub from route-filter 10.200.0.0/16 orlonger
set policy-options policy-statement HUB-FROM-OVERLAY term hub then community add HUB-LOCAL
```

```
set policy-options policy-statement HUB-FROM-OVERLAY term hub then accept
set policy-options policy-statement HUB-FROM-OVERLAY term end then reject
# to the peer: own default, tenant /24s only from the SRG1-active node, own hub-site routes
set policy-options policy-statement WAN-EXPORT term default from route-filter 0.0.0.0/0 exact
set policy-options policy-statement WAN-EXPORT term default then accept
set policy-options policy-statement WAN-EXPORT term lan from protocol static
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.10.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.20.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.30.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.40.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.50.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from route-filter 10.10.60.0/24 exact
set policy-options policy-statement WAN-EXPORT term lan from condition MNHA-ACTIVE
set policy-options policy-statement WAN-EXPORT term lan then accept
set policy-options policy-statement WAN-EXPORT term hub from community HUB-LOCAL
set policy-options policy-statement WAN-EXPORT term hub then accept
set policy-options policy-statement WAN-EXPORT term end then reject
# from the peer: hub-site routes rank below this node's own overlay routes (BGP 170)
set policy-options policy-statement ICD-IMPORT term hub from route-filter 10.200.0.0/16 orlonger
set policy-options policy-statement ICD-IMPORT term hub then preference 175
set policy-options policy-statement ICD-IMPORT term hub then accept
set policy-options policy-statement ICD-IMPORT term rest then accept
```

eBGP over ge-0/0/5 (AS 65002 to 65001, BFD 300 ms × 3) and the leak into the tenant VRs (node 2)

```
set routing-instances WAN-VR protocols bgp group ICD type external
set routing-instances WAN-VR protocols bgp group ICD family inet unicast rib-group WAN-LEAK
set routing-instances WAN-VR protocols bgp group ICD import ICD-IMPORT
set routing-instances WAN-VR protocols bgp group ICD export WAN-EXPORT
set routing-instances WAN-VR protocols bgp group ICD peer-as 65001
set routing-instances WAN-VR protocols bgp group ICD neighbor 10.255.2.1
set routing-instances WAN-VR protocols bgp group ICD bfd-liveness-detection minimum-interval 300
set routing-instances WAN-VR protocols bgp group ICD bfd-liveness-detection multiplier 3
# WAN-LEAK: every route from the peer into WAN-VR; only the peer's 0/0 into lan and GUEST-VR
set routing-options rib-groups WAN-LEAK import-rib WAN-VR.inet.0
set routing-options rib-groups WAN-LEAK import-rib lan.inet.0
set routing-options rib-groups WAN-LEAK import-rib GUEST-VR.inet.0
set routing-options rib-groups WAN-LEAK import-policy WAN-LEAK-POL
set policy-options policy-statement WAN-LEAK-POL term wan to rib WAN-VR.inet.0
set policy-options policy-statement WAN-LEAK-POL term wan then accept
set policy-options policy-statement WAN-LEAK-POL term default from route-filter 0.0.0.0/0 exact
set policy-options policy-statement WAN-LEAK-POL term default then accept
set policy-options policy-statement WAN-LEAK-POL term end then reject
```

Tenant routing instances (node 2)

```
# lan: keep Mist's overlay import, drop Mist's inet.0 default import (in the lab, inet.0 holds the out-of-band default)
set routing-instances lan routing-options instance-import vpn_OrgOverlay_bgp
delete groups top routing-instances lan routing-options instance-import wan_default
set routing-instances lan routing-options static route 0.0.0.0/0 next-table WAN-VR.inet.0
set routing-instances lan routing-options static route 0.0.0.0/0 preference 200
# GUEST-VR: GUEST and its DHCP group and pool move out of Mist's lan VR (ALQ: section 5.5)
delete groups top routing-instances lan interface ge-0/0/3.30
set routing-instances GUEST-VR instance-type virtual-router
set routing-instances GUEST-VR interface ge-0/0/3.30
delete groups top routing-instances lan system services dhcp-local-server group GUEST
delete groups top routing-instances lan access address-assignment pool GUEST
set routing-instances GUEST-VR access address-assignment pool GUEST family inet dhcp-attributes name-server 1.1.1.1
set routing-instances GUEST-VR access address-assignment pool GUEST family inet dhcp-attributes name-server 8.8.8.8
set routing-instances GUEST-VR access address-assignment pool GUEST family inet dhcp-attributes router 10.10.30.1
set routing-instances GUEST-VR access address-assignment pool GUEST family inet network 10.10.30.0/24
set routing-instances GUEST-VR access address-assignment pool GUEST family inet range GUEST high 10.10.30.200
set routing-instances GUEST-VR access address-assignment pool GUEST family inet range GUEST low 10.10.30.10
set routing-instances GUEST-VR system services dhcp-local-server group GUEST interface ge-0/0/3.30
set routing-instances GUEST-VR routing-options static route 0.0.0.0/0 next-table WAN-VR.inet.0
# no APBR: VR routing decides the exit
delete apply-groups apbr
```

Zone ICD and the tenant → ICD policies (identical on both nodes)

```
set security zones security-zone ICD interfaces ge-0/0/5.0 host-inbound-traffic system-services all
set security zones security-zone ICD interfaces ge-0/0/5.0 host-inbound-traffic protocols all
# ingress node: one tenant -> ICD policy per tenant zone
set security policies from-zone MGMT to-zone ICD policy MGMT-xnode match source-address any
```

```
set security policies from-zone MGMT to-zone ICD policy MGMT-xnode match destination-address any
set security policies from-zone MGMT to-zone ICD policy MGMT-xnode match application any
set security policies from-zone MGMT to-zone ICD policy MGMT-xnode then permit
set security policies from-zone CORP to-zone ICD policy CORP-xnode match source-address any
set security policies from-zone CORP to-zone ICD policy CORP-xnode match destination-address any
set security policies from-zone CORP to-zone ICD policy CORP-xnode match application any
set security policies from-zone CORP to-zone ICD policy CORP-xnode then permit
set security policies from-zone GUEST to-zone ICD policy GUEST-xnode match source-address any
set security policies from-zone GUEST to-zone ICD policy GUEST-xnode match destination-address any
set security policies from-zone GUEST to-zone ICD policy GUEST-xnode match application any
set security policies from-zone GUEST to-zone ICD policy GUEST-xnode then permit
set security policies from-zone MEDIA to-zone ICD policy MEDIA-xnode match source-address any
set security policies from-zone MEDIA to-zone ICD policy MEDIA-xnode match destination-address any
set security policies from-zone MEDIA to-zone ICD policy MEDIA-xnode match application any
set security policies from-zone MEDIA to-zone ICD policy MEDIA-xnode then permit
set security policies from-zone SECURITY to-zone ICD policy SECURITY-xnode match source-address any
set security policies from-zone SECURITY to-zone ICD policy SECURITY-xnode match destination-address any
set security policies from-zone SECURITY to-zone ICD policy SECURITY-xnode match application any
set security policies from-zone SECURITY to-zone ICD policy SECURITY-xnode then permit
# the GEO-BLOCK-DE lines (section 5.1) go here, above IOT-xnode, not at the end of the list
set security policies from-zone IOT to-zone ICD policy IOT-xnode match source-address any
set security policies from-zone IOT to-zone ICD policy IOT-xnode match destination-address any
set security policies from-zone IOT to-zone ICD policy IOT-xnode match application any
set security policies from-zone IOT to-zone ICD policy IOT-xnode then permit
```

ICD → local WAN and ICD → overlay (node 2, egress node)

```
# from the peer to this node's ISP (NAT: ICD_to_wan2, section 5.3)
set security policies from-zone ICD to-zone wan2 policy xnode-out match source-address any
set security policies from-zone ICD to-zone wan2 policy xnode-out match destination-address any
set security policies from-zone ICD to-zone wan2 policy xnode-out match application any
set security policies from-zone ICD to-zone wan2 policy xnode-out then permit
# from the peer into this node's tunnel to the hub
set security policies from-zone ICD to-zone vpn_OrgOverlay policy xnode-iot match source-address any
set security policies from-zone ICD to-zone vpn_OrgOverlay policy xnode-iot match destination-address any
set security policies from-zone ICD to-zone vpn_OrgOverlay policy xnode-iot match application any
set security policies from-zone ICD to-zone vpn_OrgOverlay policy xnode-iot then permit
```

Node 1 differences

```
set interfaces ge-0/0/5 unit 0 family inet address 10.255.2.1/30
set routing-instances WAN-VR routing-options router-id 10.255.2.1
set routing-instances WAN-VR routing-options autonomous-system 65001
set routing-instances WAN-VR protocols bgp group ICD peer-as 65002
set routing-instances WAN-VR protocols bgp group ICD neighbor 10.255.2.2
# lan: no preference line (ISP 1 is local)
set routing-instances lan routing-options static route 0.0.0.0/0 next-table WAN-VR.inet.0
# GUEST-VR: the local default sits behind node 2's BGP 0/0
set routing-instances GUEST-VR routing-options static route 0.0.0.0/0 next-table WAN-VR.inet.0
set routing-instances GUEST-VR routing-options static route 0.0.0.0/0 preference 200
# WAN zone is wan
set security policies from-zone ICD to-zone wan policy xnode-out match source-address any
set security policies from-zone ICD to-zone wan policy xnode-out match destination-address any
set security policies from-zone ICD to-zone wan policy xnode-out match application any
set security policies from-zone ICD to-zone wan policy xnode-out then permit
```

APBR. Mist renders `groups apbr` from the application policies. The `delete apply-groups apbr` line keeps that group from being applied; keep it in both profiles. To check, run `show configuration apply-groups` and confirm that `apbr` is not listed.

WAN-EXPORT. Term `default` exports the node's ISP default (the DHCP route in `WAN-VR`). It has no probe of the upstream path. Term `lan` uses the active-signal-route condition `MNHA-ACTIVE` (section 4.4). Only the SRG1-active node advertises the tenant /24s, so the peer's `WAN-VR` sends cross-node return traffic back to the active node. Term `hub` sends only the routes from the node's own tunnel.

As built, the cross-node policies match any source, destination and application, and zone `ICD` accepts all host-inbound services and protocols. Scoping both items, and per-tenant Internet behavior on ISP or ge-0/0/5 failure, are not yet tested (section 9).

5.5 DHCP server with Active Leasequery (node 2 shown)

Mist renders the DHCP groups and pools from each network's DHCP settings in the Hub Profile: range .10 to .200, gateway = the SRG1 virtual IP (.1). It renders them in routing instance `lan`, inside configuration group `top`. The lines below are node 2's Additional CLI as built (15 Oct), plus the two D30.2 reply lines at the end, which are not in the profiles.

```
# SRG1-active node answers DHCP; clients get the SRG1 virtual IP as gateway (one virtual IP per tenant network)
set chassis high-availability services-redundancy-group 1 managed-services dhcp
# SRG1 virtual IPs 1-6 (tenant gateways): section 4.4

# GUEST DHCP group and pool in GUEST-VR: section 5.4 (Tenant routing instances)

# Active Leasequery: one peer per DHCP routing instance = the other node's address in that instance
set routing-instances lan system services dhcp-local-server active-leasequery peer-address 10.10.10.2
set routing-instances lan system services dhcp-local-server active-leasequery timeout 60
set routing-instances lan system services dhcp-local-server active-leasequery idle-timeout 10
set routing-instances GUEST-VR system services dhcp-local-server active-leasequery peer-address 10.10.30.2
set routing-instances GUEST-VR system services dhcp-local-server active-leasequery timeout 60
set routing-instances GUEST-VR system services dhcp-local-server active-leasequery idle-timeout 10
set system processes dhcp-service accept-max-tcp-connections 100
set system processes dhcp-service request-max-tcp-connections 100
set system processes dhcp-service max-leasequery-reply-packet-retries 40
set system processes dhcp-service leasequery-reply-packet-retry-timer 20
set system processes dhcp-service max-leasequery-reply-packet-process 500

# Required on D30.2: OFFER and ACK sent as broadcast. NOT in the 15 Oct profiles; add to both nodes.
# Routing-instance form: one line per DHCP routing instance (lan and GUEST-VR)
set routing-instances lan system services dhcp-local-server overrides no-unicast-replies
set routing-instances GUEST-VR system services dhcp-local-server overrides no-unicast-replies
```

Per-node value	Node 1	Node 2
Own address on MGMT / GUEST (Mist IP config)	10.10.10.2 / 10.10.30.2	10.10.10.3 / 10.10.30.3
ALQ peer in `lan`	10.10.10.3	10.10.10.2
ALQ peer in `GUEST-VR`	10.10.30.3	10.10.30.2

The `managed-services dhcp` and `dhcp-service` lines are the same on both nodes. The GUEST group and pool are removed from Mist's render and re-created in `GUEST-VR` with fixed values (section 5.4). A later change to the GUEST DHCP settings in a profile does not reach `GUEST-VR`: edit the `GUEST-VR` lines in both profiles and in node 2's device-level list (section 4.2). Mist renders host-inbound `system-services dhcp` on each LAN zone, which is enough for the ALQ session (TCP/67). Check both nodes with `show dhcp server active-leasequery summary routing-instance lan` and `show dhcp server active-leasequery summary routing-instance GUEST-VR`: one peer, Connected. An unreachable peer is not listed at all.

no-unicast-replies, test status. Routing-instance form: tested by CLI in `lan` on node 1, D30.2, 14 Oct, with a Linux client: with the line the client binds; without it the client receives no OFFER. The system-level form, `set system services dhcp-local-server overrides no-unicast-replies`, covers only DHCP groups in the default instance; this design has none there. It was tested on D30.2 on the WAN Edge Template build (section 2), whose DHCP groups were in the default instance (16 Oct). Not yet tested (section 9): the lines pushed from the Hub Profiles, and the `GUEST-VR` line by any method. Carry both lines in the Additional CLI of both profiles. An on-box copy is removed by a later Mist push (section 4.1).

The DHCP sync results in section 3.6 and section 6.2 are from 5 Oct 2026, 25.4X300-D30. That run used an earlier revision of these lines; section 3.6 gives the `lan` ALQ peer used in each run.

5.6 IPsec to the hub (node 2 shown)

Mist renders the overlay from the profile settings below. The Additional CLI lines after the table are added on top of the render (as built, node 2).

Mist setting	Value	What Mist renders
WAN port overlay path	Spoke, BFD profile broadband (section 4.3)	Certificate-based IKEv2 gateway `<wan>_to_<hub-MAC>_wan` and IPsec VPN to the hub's WAN address; multipoint `st0.0` (100.65.0.x/18); routing instance and zone `vpn_OrgOverlay` with `st0.0` and the `lo0.1` overlay loopback (100.101.0.x); iBGP group `OrgOverlay_overlay` to the hub's overlay loopback, reached over `st0.0`
Network IOT, VPN access	OrgOverlay, routed	Export policy `OrgOverlay_advertise` with the IOT subnet 10.10.60.0/24
Application policy **allow-to-hub**	IOT → hub-site networks, via OrgOverlay	IOT → `vpn_OrgOverlay` permit

```
# overlay iBGP runs the org overlay AS; the eBGP between the nodes uses WAN-VR's own AS (section 5.4)
set routing-instances vpn_OrgOverlay routing-options autonomous-system 65000

# node-local: each node negotiates its own tunnel; the wildcard matches the Mist-generated gateway name
set groups MNHA-NODE-LOCAL security ike gateway <*> node-local
set apply-groups MNHA-NODE-LOCAL

# BFD on the overlay iBGP (BGP hold time is 90 s)
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay bfd-liveness-detection minimum-interval 1000
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay bfd-liveness-detection multiplier 3

# hub prefers the SRG1-active node's IOT route (local-preference 200); the backup still advertises it at 100
set policy-options condition MNHA-ACTIVE if-route-exists address-family inet 10.255.3.1/32
set policy-options condition MNHA-ACTIVE if-route-exists address-family inet table inet.0
set policy-options policy-statement IOT-LP term active from condition MNHA-ACTIVE
set policy-options policy-statement IOT-LP term active then local-preference 200
set routing-instances vpn_OrgOverlay protocols bgp group OrgOverlay_overlay export [ IOT-LP OrgOverlay_advertise ]

# DPD always-send on the Mist-generated gateway (node 2: WAN port wan2)
delete security ike gateway wan2_to_<hub-MAC>_wan dead-peer-detection optimized
set security ike gateway wan2_to_<hub-MAC>_wan dead-peer-detection always-send
```

Mist generates the instance, group and policy names from the VPN name (OrgOverlay). The overlay instance must run AS 65000, the org overlay AS that the hub uses. The eBGP between the nodes runs in WAN-VR with its own AS: 65001 on node 1, 65002 on node 2. BGP BFD (1 s × 3) comes from these lines. The BFD profile on the overlay path is a separate Mist path setting.

Master-instance AS. As built, each profile also sets `routing-options autonomous-system` in the master instance to the node's WAN-VR AS (65001 on node 1, 65002 on node 2). This replaces the AS 65000 that Mist renders there, so the overlay instance sets AS 65000 itself. No BGP runs in the master instance in this design. A build without the master-instance line has not been tested.

`IOT-LP` is first in the export chain. It has no accept or reject action. It sets local-preference 200 only while the SRG1 active signal route 10.255.3.1 is in inet.0, which is true only on the active node. `OrgOverlay_advertise` then exports the IOT subnet. VR `lan` takes the hub-site routes from `vpn_OrgOverlay` through Mist's instance-import `vpn_OrgOverlay_bgp`.

Mist builds the IKE gateway name for each node from that node's WAN port name, the hub's MAC address and the hub's WAN port name: `wan2_to_<hub-MAC>_wan` on node 2 and `wan_to_<hub-MAC>_wan` on node 1. Copy it from each node's render. Check it again after a WAN port rename or a hub change. A wildcard group does not remove Mist's rendered `dead-peer-detection optimized`, so these lines name the gateway. Node 1 is identical except for the gateway name. Mist sets each node's `st0.0` address and `lo0.1` loopback.

5.7 Static over IPsec (alternative to 5.6 BGP)

LAB ONLY: these static routes were added on the box for the test in section 6.4 and removed afterwards. They are not in the Hub Profiles, which carry BGP over IPsec only.

```
# both SRX400 nodes: hub-site test host via the hub tunnel address
set routing-instances vpn_OrgOverlay routing-options static route 10.200.8.1/32 next-hop 100.65.0.1

# hub: tenant subnet via node 1's tunnel (100.65.0.2), floating static via node 2's tunnel (100.65.0.3)
set routing-instances vpn_OrgOverlay routing-options static route 10.10.60.0/24 qualified-next-hop 100.65.0.2 preference 5
set routing-instances vpn_OrgOverlay routing-options static route 10.10.60.0/24 qualified-next-hop 100.65.0.3 preference 10
```

Overlay BGP stayed up during the 6.4 test. VR `lan` imports only BGP routes from `vpn_OrgOverlay` (Mist's `vpn_OrgOverlay_bgp`), so the 6.4 static result measures the hub's primary and floating static return path. Static-only operation has not been tested (section 9).

6. Results

6.1 GeoIP with live traffic, before and after failover

Tested on the Hub Profile build, 15 Oct 2026, both nodes 25.4X300-D30. Test host: Linux client on the IOT VLAN, address 10.10.60.63 from the SRX DHCP server.

Test setup (LAB ONLY): the GEO-BLOCK-DE policies on both nodes and a source-NAT pool on node 1 were added on the box for this test, with a temporary Internet uplink on ISP 1. All three were removed afterwards. The pool replaced interface NAT on node 1's `CORP_to_wan`, `IOT_to_wan` and `ICD_to_wan` rule-sets. Node 1 had the deny on its WAN path only; node 2 had it on its WAN and ICD paths. Internet access for the test was through ISP 1 only. The GEO-DE object was added on the box when the nodes were enrolled, before this test, and was kept. The GeoIP lines in the Hub Profiles (section 5.1), GEO-DE included, were added after this test.

Active node	Test	Result	Evidence
Node 2	8.8.8.8, google.com – **United States (US)**	**ALLOWED**	Replies 14–36 ms; permitted, NAT on the egress node (node 1)
Node 2	193.99.144.80 – **Germany (DE)**	**BLOCKED**	GEO-BLOCK-DE on node 2, IOT → ICD: 3 lookups, 3 packets in, 0 out
SRG1 failover: node 2 → node 1			
Node 1	google.com – **United States (US)**	**ALLOWED**	Replies ~36 ms; NAT pool translations climbing
Node 1	193.99.144.80 – **Germany (DE)**	**BLOCKED**	GEO-BLOCK-DE on node 1, IOT → wan: 3 hits
Node 1	Client DHCP address	**KEPT**	10.10.60.63 before and after one failover; no renew was recorded, so this does not show lease sync

```
show security policies hit-count from-zone IOT to-zone ICD          (node 2, before failover; excerpt)
  IOT   ICD   GEO-BLOCK-DE      3   Deny

show security policies hit-count from-zone IOT to-zone wan          (node 1, after failover)
  IOT   wan   GEO-BLOCK-DE      3   Deny
  IOT   wan   01_Internet     752   Permit
```

In this test the IOT tenant's Internet egress was node 1 (ISP 1) in both states. With node 2 active, node 2 sent the traffic across the transit link (zone ICD), so its block counted on IOT → ICD. With node 1 active, the block counted on IOT → wan.

6.2 Enrollment, feeds and DHCP

Enrollment and feeds: Hub Profile build, 15 Oct 2026, both nodes 25.4X300-D30. SecIntel via fxp0: node 1 only, as an on-box test change that was removed afterwards (LAB ONLY). DHCP: an earlier configuration of the same Hub Profiles, 5 Oct 2026, both nodes 25.4X300-D30; the DHCP client was an EX2300 switch on CORP. The `lan` ALQ peer used in each run is given in section 3.6.

Test	Result	Evidence
Enrollment, both nodes	**ENROLLED**	"Device enrolled successfully!"; license type premium; device ID = chassis serial
GeoIP feed	**LOADED**	GEO-DE 31,029 IPv4 / 27,638 IPv6 entries on both nodes
Backup, never active	**REFRESHING**	Full GeoIP download and store; periodic checks every ~1 min
Node 2: active → backup	**REFRESHING**	Periodic checks continued; manual download fetched the manifest
Node 1: active → backup	**REFRESHING**	Periodic checks continued; ATP Cloud Connected throughout
SecIntel via fxp0 (`mgmt_junos`), node 1	**DOWNLOADED**	Manifest and feeds checked through the management interface
DHCP lease on both nodes (5 Oct)	**SYNCED**	Same IP and expiry on both nodes (ALQ peer Connected)
DHCP renew after failover / failback (5 Oct)	**SAME IP**	ACK with the same address, both directions
DHCP after active node reboot (5 Oct)	**SAME IP**	Survivor kept serving; rebooted node re-learned bindings over ALQ

6.3 IPsec to the hub, before and after failover

Tested on the Hub Profile build, 15 Oct 2026, both nodes 25.4X300-D30.

Active node	Test	Result	Evidence
Both	IKE / IPsec / BGP to the hub	**UP**	IKEv2 UP, 1 IPsec SA (AES-256-GCM) and BGP Established on each node
Node 2	Ping 10.200.8.1 (hub site)	**5/5**	~4 ms; 5 hits on the IOT → overlay policy on node 2
SRG1 failover: node 2 → node 1			
Hub	Return route to the IOT subnet	**MOVED**	Best path changed to node 1 (local-preference 200) about 1 s after failover

Node 1	Ping 10.200.8.1 (hub site)	**5/5**	~4 ms; 5 hits on the IOT → overlay policy on node 1

```
hub> show route 10.10.60.0/24 table vpn_OrgOverlay.inet.0        (after failover)
10.10.60.0/24  *[BGP/170] localpref 200, from 100.101.0.1 (node 1)  via st0.0
                [BGP/170] localpref 100, from 100.101.0.2 (node 2)  via st0.0
```

`vpn_OrgOverlay` is the routing instance Mist generates for the OrgOverlay VPN, on the hub and on both nodes.

6.4 ISP 1 failure: BGP over IPsec vs static over IPsec

Tested on the Hub Profile build, 15 Oct 2026, both nodes 25.4X300-D30. Node 1 active (its tunnel uses ISP 1). Continuous ping from the IOT host to the hub site (10.200.8.1) every 0.5 s; ISP 1 disconnected, then restored.

Static over IPsec column (LAB ONLY): the static routes on both nodes and on the hub (section 5.7) were added on the box for this test and removed afterwards. They are not in the Hub Profiles.

Behavior	BGP over IPsec (BFD 1 s × 3)	Static over IPsec (primary + floating)
Hub return path	Node 1 route withdrawn immediately; node 2 path used	Node 1 tunnel next hop removed by dead-peer detection; floating static to node 2 used
Node 1 outbound path	Across the transit link (ge-0/0/5), out node 2's tunnel	Across the transit link (ge-0/0/5), out node 2's tunnel
SRG1	Stayed on node 1	Stayed on node 1
Packet loss (failure + restore)	**5 of 402 (~2.5 s)**	**5 of 298 (~2.5 s)**
After ISP 1 restored	Tunnel and BGP re-established; hub back to node 1 (local-preference 200) automatically	Tunnel re-established ~23 s later; hub back to the primary static automatically
Follows an SRG1 failover	Yes — preference set from SRG1 state	No — preference fixed in configuration

6.3 and 6.4 used ping, which opens a new session for each echo. Established sessions to the hub site were not tested across an SRG1 failover, an ISP failure or the ISP restore. On ISP 1 failure, node 1's egress for hub-site traffic moves from st0.0 (zone `vpn_OrgOverlay`) to ge-0/0/5 (zone `ICD`). ISP 2 failure with node 2 active was not tested (section 9).

6.5 LAN link failure on the active node

Tested on the WAN Edge Template build, 19 Oct 2026, both nodes 25.4X300-D30 (section 2, Test builds). This is a separate build of the same pair, with Mist's native MNHA settings in a WAN Edge Template and per-node values set through the Mist API. It differs from the Hub Profile build of sections 4 and 5 in the ICL, the liveness multiplier, the SRG layout, the routing and the NAT. This test has not been repeated on the Hub Profile build.

Node 1 active. Continuous ping from the IOT host to the hub site (10.200.8.1) every 0.5 s. The access-switch port facing node 1's LAN interface was disabled for about 90 s, then re-enabled; SRG1 was then moved back to node 1 by command.

Time	Event	Evidence
0 s	Node 1 LAN link down; node 1 leaves SRG1	Node 1 `ACTIVE → INELIGIBLE, Reason: Interface monitoring down`
+0.2 s	Node 2 takes SRG1	Node 2 `BACKUP → ACTIVE, Reason: Peer in Ineligible state`; IOT virtual IP on node 2's LAN unit
+1 s	Hub return path moves	Hub learns the IOT subnet from node 2 at local-preference 200; node 1's route withdrawn within 3 s
Link restored	Node 1 rejoins	Node 1 `INELIGIBLE → BACKUP, Reason: Local node recovered`; cold sync COMPLETE; SRG1 stays on node 2 (preemption disabled)
Failback	Manual SRG1 failover to node 1	Node 1 `BACKUP → ACTIVE`; hub back to node 1 (local-preference 200) in the same second
Packet loss, whole sequence		**0 of 626**

Only the ping to the hub site was measured. Internet traffic was not measured in this test (section 9).

7. Verification checklist

Run each check on both nodes unless the row says otherwise.

Check	Command	Expect
MNHA state	`show chassis high-availability information`	Peer on ge-0/0/4.0: Conn State UP, Cold Sync Status COMPLETE; Peer Forwarding IP on ge-0/0/4.0 with Peer ICD Conn State UP; SRG1 ACTIVE on one node, BACKUP on the other; Preemption DISABLED; Failure Events: NONE
LAN monitoring	`show chassis high-availability services-redundancy-group 1`	Interface Monitoring: `ge-0/0/3` State Up, on both nodes
MNHA config	`show configuration chassis high-availability | display inheritance no-comments | display set | no-more`	The `chassis high-availability` lines of the node's own profile (section 4.4), and no others. A `chassis high-availability` line removed from the profile stays on the box (D30.2) until it is deleted there by CLI
Session sync	`show chassis high-availability data-plane statistics`	Session create: RTOs received on the backup rise with RTOs sent on the active node. Compare the change over a minute, not the totals, which reset when a node reboots
Sessions on both nodes	`show security flow session summary`	Similar valid-session count on both nodes
Transit link BGP	`show bgp summary instance WAN-VR`	Group ICD peer Established: 10.255.2.2 on node 1, 10.255.2.1 on node 2

Transit link routes	`show route receive-protocol bgp <peer> table WAN-VR.inet.0 | no-more`	From the peer (10.255.2.2 on node 1, 10.255.2.1 on node 2): 0/0 while the peer's ISP is up, and the hub-site prefixes while the peer's tunnel is up. The six tenant /24s arrive only on the backup node, from the SRG1-active peer
Tenant egress	`show route 0.0.0.0/0 exact table lan.inet.0 | no-more` `show route 0.0.0.0/0 exact table GUEST-VR.inet.0 | no-more`	With both ISPs up. lan: Static (to table WAN-VR.inet.0) active on node 1; BGP via 10.255.2.1 active on node 2. GUEST-VR: BGP via 10.255.2.2 active on node 1; Static (to table WAN-VR.inet.0) active on node 2
APBR removed	`show configuration apply-groups`	`apbr` not listed
BFD	`show bfd session`	All Up. Transit link peer on ge-0/0/5.0: 300 ms × 3. Hub overlay neighbor: 1000 ms × 3. MNHA ICL and forwarding IP (10.255.0.x, 10.255.1.x): 1000 ms × 3
IPsec	`show security ike security-associations` `show security ipsec security-associations`	UP on both nodes
Overlay BGP	`show bgp summary instance vpn_OrgOverlay`	Hub neighbor Established on both nodes
Signal route on the active node only	`show route 10.255.3.1/32 exact`	Active node: 10.255.3.1/32 present (the route the MNHA-ACTIVE condition tests). Backup: no 10.255.3.1/32 (it holds 10.255.3.2/32)
Hub return path	hub: `show route 10.10.60.0/24 table vpn_OrgOverlay.inet.0`	Best path = active node (local-preference 200); backup node at 100
Cloud connection	`show services advanced-anti-malware status`	Connected on both nodes
Feed refresh	`show services security-intelligence update status`	succeeded; time advances on both nodes
GeoIP object	`show security dynamic-address summary`	GEO-DE IPv4/IPv6 entries > 0
GeoIP policy order	`show security policies from-zone IOT to-zone wan` (node 1) `show security policies from-zone IOT to-zone wan2` (node 2) `show security policies from-zone IOT to-zone ICD`	GEO-BLOCK-DE listed first (sequence 1), above 01_Internet or IOT-xnode. Recheck after any Mist application-policy change
Policy enforced	`show security policies hit-count`	GEO-BLOCK-DE increases on the active node: IOT → wan on node 1; IOT → ICD on node 2 (its lan tenants exit ISP 1 through node 1)

NAT	`show security nat source rule all`	Translation hits increase on the egress node: `<tenant>_to_wan` (node 1) or `<tenant>_to_wan2` (node 2) for local traffic; `ICD_to_wan` / `ICD_to_wan2` for traffic from the peer
DHCP sync	`show dhcp server active-leasequery summary routing-instance <ri>`	Peer Connected for lan and for GUEST-VR, on both nodes. A missing peer line counts as a failure
DHCP bindings	`show dhcp server binding routing-instance <ri>`	Same leases on both nodes; equal binding count per routing instance
Mist commit	`show system commit`	A new commit by Mist after each profile change on that node. Check this before acting on the portal status: a commit shown as FAILED can later show COMMITTED, for example after a retry
Manual failover / failback	`request chassis high-availability failover services-redundancy-group 1 peer-id <target>` (on the active node)	Run when the target node shows Cold Sync COMPLETE, ALQ peer Connected, equal binding counts, and overlay BGP and BFD Up. Preemption is disabled, so failback is manual

`vpn_OrgOverlay` and `OrgOverlay_overlay` are the names Mist generates for the overlay named OrgOverlay. WAN-VR is the LAB ONLY placement of the WAN (section 9). In `show chassis high-availability information`, Peer ICD is the MNHA forwarding path on the ICL (ge-0/0/4.0). The transit link on ge-0/0/5 does not appear there; check it with the transit link BGP and BFD rows.

8. Deployment order

0. **Prerequisites.** In Mist, create the OrgOverlay VPN, the org networks MGMT, CORP, GUEST, MEDIA, SECURITY and IOT, and the HUB-SITE-NETS application that allow-to-hub uses. Build the vSRX hub's Hub Profile (hub role, static WAN; section 4.5), assign it to the hub, and confirm that the hub is connected. On the LAN switch, carry VLAN 10 (MGMT) untagged and VLANs 20, 30, 40, 50 and 60 tagged on the trunk to ge-0/0/3 of each node. The virtual IPs and both ALQ peers need these VLANs on both nodes (section 3.6).
1. In Mist, create one Hub Profile per SRX400 node (not a WAN Edge Template) and assign each to its device (section 4.2). Keep the Additional CLI in the profile. A device-level Additional CLI list replaces the profile's list; if a node has one, edit both. After every change, confirm the Mist commit on the node (`show system commit`).
2. Set the native fields in each profile: WAN port, LAN trunk with the node's own address on each tenant network, DHCP pools, IOT VPN access, the overlay spoke path, the traffic steering paths wan and OrgOverlay, and the application policies (section 4.3).
3. Add MNHA to each profile's Additional CLI: the ICL /30 on ge-0/0/4, the lo0.0 forwarding address with a static route to the peer's, MNHA forwarding on ge-0/0/4.0, zone HA, and SRG1 with virtual IPs, signal routes and `monitor interface ge-0/0/3` (section 4.4). Check the MNHA state and LAN monitoring rows (section 7).
4. Add the cross-node routing and policy (section 5.4): WAN-VR holding the WAN port and ge-0/0/5 (LAB ONLY), eBGP with BFD on the transit link (WAN-EXPORT, ICD-IMPORT, rib-group WAN-LEAK), the VR lan and GUEST-VR defaults, `delete apply-groups apbr`, zone ICD with its cross-node policies, and ICD source NAT (section 5.3). Check the transit link BGP and BFD rows.
5. Add the overlay lines (section 5.6): `vpn_OrgOverlay` autonomous-system 65000, the `MNHA-NODE-LOCAL` group, BGP BFD 1000 ms × 3, condition MNHA-ACTIVE with IOT-LP first in the export chain, and DPD

always-send on the Mist-generated gateway (`wan_to_<hub-MAC>_wan` on node 1, `wan2_to_<hub-MAC>_wan` on node 2). Add the hub's lines to its profile (section 4.5). Check the IPsec, overlay BGP, signal route and hub return path rows.

6. Add `managed-services dhcp` and an Active Leasequery peer in each DHCP routing instance, lan and GUEST-VR. On D30.2 also add `dhcp-local-server overrides no-unicast-replies` in lan and in GUEST-VR (section 5.5). Check the DHCP rows.
7. Confirm each chassis serial is entitled for ATP Cloud. Give each node a path to ATP Cloud and DNS. This lab used the out-of-band management network through the LAB ONLY inet.0 default (section 5.4); the data-plane path in this design is not yet tested (section 9).
8. Enroll each node with the ATP Cloud portal command (section 3.1). Then add the ATP Cloud / SecIntel lines to the profile (section 5.2); they reference the certificates and CA profiles that enrollment creates. Check the cloud connection and feed refresh rows.
9. Add the GeoIP object and the GEO-BLOCK-DE policies with `set` and `delete` lines only; Mist Additional CLI also accepts `#` comment lines, but not `insert` (section 5.1). In the Additional CLI list, place the IOT → ICD GEO-BLOCK-DE lines above the IOT-xnode lines from step 4; do not append them at the end of the list. The first line that creates a policy sets its position, so appended lines would leave the deny below the permit. Check the GeoIP object and policy order rows: GEO-BLOCK-DE is sequence 1 on IOT → ICD and on IOT → wan (node 1) or wan2 (node 2).
10. From a tenant host, test a permitted (US) destination, a denied (DE) destination and the hub site (10.200.8.1). Fail over with the manual failover row, repeat, then fail back.

The Mist profiles carry BGP over IPsec only. Static over IPsec (section 5.7) was a temporary on-box lab configuration and is not in the profiles.

9. Not yet tested

The design and results above do not cover the items below. Each stays open until it is tested on the SRX400 pair.

Area	As built, and what is not yet tested
GeoIP from the Hub Profiles	The 6.1 live test used on-box policies. DE blocked / US allowed with the section 5.1 lines pushed from the Hub Profiles, on both nodes and in both active states, is not tested. GEO-BLOCK-DE on tenant zones other than IOT is not tested. On the Hub Profile build, the deny on node 1 IOT → ICD (used when ISP 1 is down) and on node 2 IOT → wan2 (the local fallback) has carried no test traffic.
DHCP on the Hub Profile build	The DHCP sync results are from 5 Oct (earlier configuration, both nodes 25.4X300-D30), with a client that sets the broadcast flag. The renew-timing runs had the `lan` ALQ peer on MGMT, as built; the failover, failback, release and reboot runs had it on CORP. On the Hub Profile build (15 Oct), the IOT host kept its address across one failover, with no renew recorded during the test. Not tested on the Hub Profile build: a renew after failover, binding sync with the same expiry on both nodes, `no-unicast-replies` pushed from the Hub Profiles with a client that does not set the flag, the same line in GUEST-VR, and failback onto a node rebooted a few minutes earlier.
DHCP client routes after failover	On the WAN Edge Template build (section 2), a node that receives a client's binding over ALQ keeps that client's /32 route pointing at the other node's ALQ address after it becomes active, until the client renews. Traffic to the client still arrives, through the other node, while that node is up. In this design the `lan` ALQ peer is on MGMT. After a LAN-link failure or power-off of node 1, node 2's route for each client that node 1 last acknowledged can point at 10.10.10.2 until the client renews (T1 is about 12 h with the 24 h lease used in this lab). Internet and hub-site traffic from such a client after either failure is not tested.

Transit link exposure	Zone ICD (ge-0/0/5) allows all host-inbound services and protocols. The tenant → ICD, ICD → WAN and ICD → overlay policies permit any address, so cross-node traffic is not subject to the Mist application policies. Restricting both is not tested.
ICL loss and split-brain	No activeness-probe is configured. ICL down, ICL and transit link down together, and the MGMT VLAN (VLAN 10) not carried between the nodes are not tested. Neither is the split-brain probe that hybrid mode derives, expected toward virtual IP 1 (10.10.10.1).
Active node power-off or reboot	The 5 Oct DHCP run rebooted node 1 while it was active (software reboot, earlier configuration). States polled after the reboot command: node 2 ACTIVE at +12 s; node 1 in HOLD at +248 s, bindings back over ALQ at +351 s, BACKUP at +369 s, cold sync COMPLETE at +385 s. Not measured: a power-off, a reboot on the Hub Profile build, loss to the hub and the Internet, and the hub path move.
Established sessions	The traffic results use pings, and each ping opens a new session (6.3: 5 pings, 5 policy hits). An established TCP session, to the Internet or to the hub site, is not tested across an SRG1 failover in either direction, an ISP failure or an ISP restore. The zone pair and NAT point of a lan tenant's Internet session differ by active node. When the active node's ISP fails, its egress for an IOT session to the hub site changes from zone vpn_OrgOverlay (st0.0) to zone ICD (ge-0/0/5).
Upstream ISP failure	The ISP 1 test (node 1 active) disconnected the ISP link and measured hub traffic only. On the Hub Profile build, an ISP 2 failure with node 2 active (hub traffic across ge-0/0/5 to node 1's tunnel) is not tested. A failure beyond the link, and per-tenant Internet traffic during any ISP failure, are not tested. WAN-EXPORT advertises the default whenever the node has a DHCP default; nothing in the Additional CLI tracks the upstream path.
Transit link failure	Only ge-0/0/3 moves SRG1; ge-0/0/5 is not monitored. Loss of the transit link, alone and together with an ISP, and Internet traffic on the local fallback default (preference 200) are not tested.
Hub down	No fail-closed route covers the hub-site prefixes (10.200.0.0/16). With both tunnels down, hub-site traffic follows the tenant's default route. Hub-down behavior is not tested.
Export and import policy hardening	IOT-LP has no route-filter, the Mist-rendered OrgOverlay_advertise has no final reject, the spokes have no overlay import policy, and ICD-IMPORT ends with accept. A route-filter for the IOT /24, a trailing reject and filtered imports are not tested.
Static over IPsec	Section 6.4 measured the hub's primary and floating static return path with overlay BGP up. Static-only operation (no overlay BGP on the spokes), loss timing against the hub's DPD timers, and static behavior on SRG1 failover, LAN failure or node power-off are not tested.
Internet egress, as built	The 6.1 Internet test ran through ISP 1 only, with a temporary NAT pool on node 1 in place of interface NAT on the `CORP_to_wan`, `IOT_to_wan` and `ICD_to_wan` rule-sets. On the Hub Profile build, the as-built interface NAT has carried no test traffic, no tenant Internet traffic has left by ISP 2 (GUEST through GUEST-VR on either node; `xnode-out` and `ICD_to_wan2` on node 2), and Internet traffic from MGMT, CORP, MEDIA and SECURITY is not tested.
Source NAT pool	Per-node pool NAT, one pool per node routed by that node's ISP, is not tested.
ATP Cloud / SecIntel over the data plane	Hub Profile build: enrollment, the ATP Cloud connection, DNS and feed refresh used the out-of-band management network (LAB ONLY default in inet.0). WAN Edge Template build (section 2): both nodes enrolled, connected to ATP Cloud and downloaded the GeoIP feed through their ISP, with the WAN in inet.0. Not tested in this design: the data-plane path (with the WAN in WAN-VR, or in inet.0 as in the production build of section 5.4), including a node with its ISP down and after a failover; and SecIntel over fxp0 on a node that stays connected to Mist.

WAN in inet.0	WAN-VR is LAB ONLY. A production build keeps the WAN in inet.0 (section 5.4). A Hub Profile build with the WAN in inet.0 is not tested. The WAN Edge Template build (section 2) had the WAN in inet.0, with different routing and NAT.
Device-level Additional CLI	Removing node 2's device-level Additional CLI list, so that its profile's list applies, is not tested.
Fresh node build	A new or replacement node built from its profile in the section 8 order is not tested, including a push of the section 5.2 lines before the node is enrolled.

10. Reference configurations

Generic configurations of the Hub Profile build (15 Oct; sections 4 and 5) are published with this paper at github.com/SandmanJnpr/srx400-mnha-mist-branch. They use this paper's names and addresses; the README maps the Mist object names to the labels used here. Installation values, such as the management addresses, the hub's MAC address in the IKE gateway names and the ATP Cloud region, are placeholders in angle brackets. This is a lab-built reference by the author, not an official Juniper or HPE configuration.

File	Contents
`README.md`	How the files relate, import into Mist, placeholders, LAB ONLY items, design limits
`docs/zones-and-icd.md`	Security zones, zone policies and the transit link (zone `ICD`, eBGP in `WAN-VR`) on both nodes, with the configuration lines

`mist/org/networks.json`
`mist/org/services.json`
`mist/org/vpns.json`
	Org networks, the HUB-SITE-NETS application and the OrgOverlay VPN

`mist/hub-profiles/srx400-node1.json`
`mist/hub-profiles/srx400-node2.json`
`mist/hub-profiles/vsrx-hub.json`
	Hub Profiles of node 1, node 2 and the vSRX hub

`mist/additional-cli/srx400-node1.txt`
`mist/additional-cli/srx400-node2.txt`
`mist/additional-cli/vsrx-hub.txt`
	Each profile's Additional CLI: set/delete and # comment lines

`junos/srx400-node1.set`
`junos/srx400-node2.set`
`junos/vsrx-hub.set`
	The lab's as-built configuration of each device in display set form with the generic values substituted; adoption lines and secrets removed; derived lines listed in the README

`tools/mist_import.py`	Creates the `mist/` objects in a Mist org (Mist API)
`docs/SRX400-MNHA-Secure-Branch-Design.pdf`	This paper

Public release 1.0 (November 2026): the 20 October 2026 text with this section, generic tenant networks (also in Figures 4-4 to 4-6, new captures of a test import of node 2's profile; the Figure 4-5 path label is shown with the published name), placeholders in 5.2 and 5.4 and small wording changes (CHANGES.md). Figures 4-1 to 4-6: Mist organization name covered; 4-1 and 4-2: also MAC and management addresses, and the SRX400 software versions and version status.